Research Skills

Psychology of Education (Kibabii University)

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DIPLOMA COURSE OUTLINE FOR RESEARCH, MONITORING AND EVALUATION

1. Meaning of terms.
2. Research.
3. Monitoring.
- Evaluation.
2. Importance of research Understanding children and predicting their behaviors.
3. Improving parenting and care.
- Policy development.
1. Improving teaching and learning.

Characteristics of research.

3. Categories of research.
4. Cross-sectional.
5. Longitudinal.
- Correlational.
1. Experimental.
2. Basic research.
3. Action research.
- Applied research.
- Case studies.
1. Historical research.
2. General research methods in psychology and human learning.
3. Case studies.
4. Psychological tests.
- Clinical methods.
1. Observation
2. Survey method.
3. Experimental method.
- Self description.
- Correlational method.
1. Ethnographic method.
2. Achievement tests.
3. Rating scales.
4. Qualitative and quantitative research.
5. Definition of terms; Qualitative and quantitative research.
6. Characteristics of Qualitative and quantitative research.
- Advantages and disadvantages of qualitative and quantitative research.
- Qualitative research methods.
- Focus group discussions

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- Historical profiles.
- Case studies.
- Personal descriptions.
- Study of secondary information.
- Psychological and achievement tests.
- Quantitative research methods.
- Rating scales
- Study of secondary information.
- Participatory Learning and Action (PLA) and appreciative inquiry approaches. (AIA).
- Origin .
- Tools of PLA and AIA.

7. Designing a research study.
8. Identifying a research topic from

- Teaching experience.
- Areas of interest and expertise.
- Studying previous relevant literature.
- Verifying or negating previous research.
- Current/emerging issues in primary education.

1. Deciding on the population, sample and sampling methods.

- Selecting and developing appropriate research tools.

1. Deciding on fitting research category.
2. Deciding on data collection procedures and organization.
3. Deciding on data analysis procedures and presentation.
4. Sample and sampling methods.

- Definition of terms. Sample, population, sampling.
- Sampling methods.
- Radom/probability.

9. Data analysis and presentation procedures

- Graphs and charts.

10. Research proposal.

- Definition of term research proposal.
- Importance of research proposal.
- Format/content of a research proposal.

11. Research report.

- Format/content of research report.
- Dissemination of research findings.

12. Ethical consideration in research.

- Use of human being as subjects.
- Voluntary participation.

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- Permission to interview children.
- Risk of human participation.

13. Monitoring and evaluation.

- Importance of monitoring and evaluation.
- Monitoring and evaluation tools and methods.
- Uses of monitoring and evaluation data.

14. Basic statistics in Education.

References .

- Research, monitoring and evaluation by Ann Njenga and Margret Kabiru.
- Preliquisite skills in writing research proposals and projects by GachuigaIsaack.
- Research, monitoring and evaluation made simple by NyakwaraBegi.
- Research methods by Mugenda O. and Mugenda A.
- Research methods in ECD
- Introduction to research skills

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DEFINITION AND IMPORTANCE OF RESEARCH AND EVALUUATION

DEFINITIONS

Research is defined as:

- A diligent (well thought out) study to discover facts or to solve a problem.
- A scientific investigation carried out to discover facts or to solve a problem.
- A systematic collection of data (information) under careful defined conditions to discover facts or to solve a problem.
- A scientific investigation to produce new knowledge.

***Monitoring*:** Monitoring is defined as a systematic and continuous collection of data which is analyzed to show the progress achieved in a project over a period. This progress is measured against the objectives of the project.

Evaluation: evaluation is defined as a systematic and continuous collection of data which is analyzed and is used to make certain judgments about a project. Evaluation assesses the effects of the activities in the project on the beneficiaries (those who are meant to benefit from the project)./it is the process of making judgement of the value of a project.

CHARACTERISTICS OF RESEARCH

- ***Research is systematic*.** It follows a well structured process and has definite rules.
- ***Research is logical****.* It follows logical steps that are clearly defined from the beginning to the end.
- ***Research is empirical*.** It involves use of concrete data collected from the field or from experiments carried out.
- ***Continuous***
- ***Research is deductive*.** This means that the researcher is able to analyze and interpret data collected in order:
- To bring out better and clearer understanding of the problem being studied.
- To arrive at certain conclusions and generalizations.

IMPORTANCE OF RESEARCH IN SCHOOLS.

Research provides information which may help:

- To improve our understanding of children, how they grow, develop and learn.
- To identify the needs of children and how best to meet these needs.
- To identify services required to meet the needs of children.
- To solve problems facing children and their families.

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- To improve services for young children, from example, health services, learning environments in primary schools, growth monitoring and promotion (GMP) services e.t.c.
- To develop innovative methods of improving children's learning.
- To provide new knowledge that improves children's learning and their total wellbeing.
- Parents, teachers, communities and policy makers to appreciate the importance of nutrition in influencing growth, development and learning of young children.
- To improve the care of young children both at home and in the schools
- To identify what is required to ensure the provision of quality teaching-learning conditions in schools.
- To identify the effects of provision of needs of children in influencing growth, development and leaning.
- To improve learning-teaching processes in the schools
- To direct planning and future programmes
- To influence the development of new policies in education
- In advocacy. The research data collected can be used as a tool to convince policy makers, parents, teachers and other stakeholders about the importance of early childhood period and the need to invest in these years.

CATEGORIES OF RESEARCH AND RESEARCH DESIGNS

CATEGORIES OF RESEARCH

1. **Basic Research**

The main aim of basic research is to add new knowledge. It does not necessarily produce results of immediate practical application. It is mostly based on testing or applying theory. However, not all basic research is based on theory. A theory consists of systematically organized ideas and observable data which is used to predict or explain behavior.

2. **Applied Research**

Applied research is undertaken to solve an immediate practical problem. In action research knowledge which has been gained through theory can be applied to solve practical problems. Not all applied researches are however based on theory. Whether based on theory or not, research should be systematic and well organized in order to obtain valid data.

3. **Action Research**

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Action research is a type of applied research. Action research is used to solve specific problems. It is aimed at improving practices and is usually carried out by practitioners such as teachers and administrators. Action research uses modified basic research guidelines. Action research is supposed to improve practices, for example, provide ideas on better methods of teaching young children. It can be carried out by individuals, teams or organizations. Research undertaken by teams is sometimes referred to as ***collaborative research.***

Action research should be organized in a systematic way to minimize bias and misinterpretation Classroom teachers, schools can use action research to improve teaching-learning methodologies. This can lead to improvement in learning environments and pupil performance. Action research can also be done on a large scale, for example, during piloting and implementation of a curriculum, a lot of data is collected by teachers, education field officers and curriculum developers. This data is used to decide on areas that need to be modified or strengthened. Action research is characterized by the following features:

- It focuses on specific problems within a specific context.
- It involves planned ongoing activities and interventions which are constantly reviewed to increase understanding of the processes and factors involved.
- It is usually participatory in nature involving the practitioners (for example teachers) and target groups or beneficiaries (for example learners) in taking decisions and monitoring the interventions.
- The practitioners and the target groups engage in continuous reflection so that they can learn and creatively contribute to improving the programme or project.

Strengths

- It provides practical solutions to problems.
- It focuses on the situation that the practitioner is engaged in. it is therefore likely to be relevant and to motivate practitioners to participate fully in it.
- The participants are happy and comfortable with the research. They are not threatened by it because it deals with problems that affect their daily lives. Since the participants are happy with the research they participate in it and are likely to implement recommendations arising from the programme or project.
- The practitioners and target groups acquire new knowledge and skills which they can use to improve their work.

Limitations.

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- The researcher who is also a practitioner can be subjective while interpreting the results.
- It is difficult to balance the 'practice' and 'research' aspects.
- The practitioner may lack some necessary research skills to ensure that the action research findings are reliable.
- It might not be possible to generalize results to other situations.

Basic Research Designs

Descriptive research

Descriptive research is used for observing and recording behavior. It consists of naturalistic observation, participant observation, interviews, questionnaires, case studies, ethnographic studies and historical research.

1. ***Naturalistic observation research.***

In naturalistic research, the participants are studied in their real natural world. One can, for example, do a naturalistic research by observing children as they play in the neighborhood, or observe them in classroom.

Advantage.

- The researcher can study without interfering with the participants' activities.
- The participants are likely to show their real behavior.

Limitations.

- The study can be time consuming
- It is likely to be expensive
- Presence of the researcher can influence the behavior of the people being studied
- Researchers might disagree on what they have seen
- This type of research does not tell us about causes and effects of what is observed.

2. ***Participation***

Participation observation occurs when the observer or researcher is actively involved as a participant in the research activity or intervention. The researcher takes time to observe what he/she researching on and the rest of the time participates in the activities that the people he/she is studying are involved in. the researcher observes whatever he is supposed to observe over a period of time. He takes detailed notes. The information collected help to reveal the progress made, patterns and trends.

Strengths

- The researcher can obtain a lot of data as those participation in the activity might be free in providing information.

Limitations

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- The researcher could become less objective if he/she becomes part of the implementing team.

3. ***Interviews and questionnaires***

Interviews and questionnaires are also used to obtain descriptive data.

4. ***Case studies***

A case study is an-depth look at an individual or a single entity. A case study of a person can be done by tracing the case history from birth and noting the experiences and environments that the person has been through. Several methods such as published biographical materials, observation, clinical interviews, medical examination and tests, or achievement tests can be used to collect data for case studies. Such case studies are helpful for persons in need of guidance and counseling, medical treatment and rehabilitation. The case study method can also be used of study groups or projects. It can provide data on the history of the group, group dynamics and processes.

Advantages

- Case studies provide in-depth information about an individual.
- The results can be fairly accurate if the observations and other data collection methods are carried out well.

Limitations

- Case studies are time consuming and very expensive.
- Verification of information can be difficult and cannot be generalized to other people or situations as each person or situation is unique.
- The case study data can be unreliable if the person collecting data is biased. Often case study data is not verified by other persons.

5. ***Ethnographic studies***

An ethnographic study is an in-depth description and interpretation of behaviours in a cultural group. The researcher lives among the group, participates in the group activities and collects information through interviews, observation and study of documents.

Strengths

- It is possible to collect in depth data about people's lives.

Limitations

- It is time consuming and very expensive
- The researcher can provide biased data which are not easily verifiable by other researchers
- The data are not generalizable to other situations.

6. ***Historical Research***

Historical research is the systematic and objective study which evaluates and synthesizes data collected in order to establish facts and draw conclusions

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concerning past events. In order to obtain credible facts and carry out a good historical research it is important to ensure that:

- You have clear goal, objectives or hypothesis to ensure that the study is focuses.
- You have sufficient evidence in the area of interest before embarking on the research.
- You know where the source of materials are and have a plan how they will be accessed.
- You are able to assess the facts for accuracy and authenticity through internal and external criticism. Internal criticism checks on the accuracy of the facts. The external criticism establishes whether the documents and sources are genuine. It establishes who the author of the source documents was, whether he/ she was a trained historian, why the document was written and under what circumstances.
- After collecting the information, you as the researcher sorts out the facts, synthesizes them and tries to identify trends and patterns, by so doing you are in a better position to balance the facts and to give correct interpretation.

Strengths

- It helps us to understand the origins of our educational institutions.
- It helps us to understand the origins of practices used in educational institutions.
- It helps us appreciate how present practices and problems have evolved.
- We can learn from past successes and avoid repeating mistakes of them past.
- We can use past experiences to predict the future.

Limitations

- Historical research cannot give conclusive results about past events.
- There is possibility of errors in recording and interpreting events that happened in the past.
- It is possible for the researcher to introduce own biases in the documentation and interpretation of the past events.
- It is time consuming and expensive.
-

Correlational Research

Correlational design is a design in which the researcher gathers information without changing the participant's experiences and examine relations between variables. Variables are factors which may influence changes in a population. Examples of variables include age, formal education level, gender, religious background and socio-economic level. Correlation refers to the degree of

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relationship between two or more variables. A correlation coefficient ranges from1.0 to 1.0. A correlation of 0 means that there is no relationship between two variables. A positive correlation indicates that the two variables tend to change in the same direction. For example, the older people get the more patient they become. A negative correlation between two variables indicates that they tend to change in the opposite direction. For example, the older one gets the slower the physical movements. When two variables are correlated, they can be used to predict one another. Two positively correlated variables can cause one another or both can be caused by another variable.
Example of positively correlated variables.

Another example, if you give 3 tests of maths, history and geography to a group of pupils. The results of the tests can be analyzed statistically to obtain correlation coefficients. If the correlation coefficient between the marks and geography is 80 while that between maths and history is 50, we can conclude that maths is likely to be a better predictor of performance in geography than in history. Correlation, however, does not mean causation. From example, if parent-child relationship correlates with high academic performance, you cannot say for certain, that high academic performance was caused by the parent-child relationship because there are many factors that influence academic performance. Also academic performance can influence parent-child relationship.

Strengths

- One can study relationship between variables.

Limitations

- Does not allow inference on cause and effect relationships.

Experimental research design

Experimental research design comprises experimental design, modified experimental design and natural experiment.

1. Experimental design

This is a research design in which participants are randomly assigned to two or more treatment conditions. In such a design, one studies the effect that manipulating an independent variable has on a dependent variable. An independent variable is the one anticipated or hypothesised to cause a change in the other variable known as the dependent variable. Most experimental researches are conducted in laboratories to allow maximum control of the variables. A treatment such as a new teaching method is introduced in one

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group and not in another similar group which is referred to as a control group. The two groups are matched for all factors except the teaching method.

Strengths

- Permits inferences about cause and effect.

Limitations

- Findings may not generalise to real world situations.
- It is not possible to do experiments in all aspects of behaviour because of ethical and practical factors.
-
- The control and experimental participants often influence one another.
- The experiment might create an unnatural environment that can influence results.

2.Modified experimental designs

Field experiments are a type of modified experimental design. Field experiments are designs in which the participants are randomly assigned to treatment conditions in natural settings. Treatment refers to a condition such-as training, teaching method or a type of reinforcement. For example, one can randomly assign children from one class into two groups. One group is taught maths by a teacher who only works all the examples on the chalk board. In the other group, the teacher works the examples on the chalk board and then children practice in their exercise books. After some lessons, children are tested to find out if there is a difference in the performance of the two groups.

Strengths

- Permits generalization of findings to the real world.

Limitations

- Control over treatment is not as good as in laboratory experiments.

3.. ***Natural experiment***

This is a design in which the researcher studies already existing treatments in natural settings by .carefully selecting groups of participants with similar characteristics. One can, for example, compare pre-schools, carefully selecting the participants to ensure they have similar characteristics. In natural experiments, one can study the same group before and after the treatment and compare it with a group that has not undergone the treatment.

Strengths

- Permits study of naturally occurring variables which are not under control of the researcher.

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Limitations

- Obtained differences may be due to other factors other than treatment.

Time-span research designs

Longitudinal design

Longitudinal design is a research in which participants are studied repeatedly at different ages over a period of time. The time span may range from months to several years. The same group of children can be studied on their performance in maths or social skills at the age of 3, 5 and 7 years.

Strengths

- One can track performance or behaviour of a person over time. This allows the researcher to see patterns as well as individual differences in development.
- It is possible to examine relationships between early and later events and behaviour.
- Longitudinal studies show stability and changes in development over time.

Limitations

- Longitudinal studies are time consuming and can be quite expensive.
- These studies take a long time. Therefore, usually few people agree to participate in such

studies.

- Biased sampling where people who participate in such studies are likely to have unique characteristics which might not represent the wider population.
- Selective attrition where many participants are likely to drop out and those who remain could be different in important ways. The results based on the remaining group cannot be generalized to the wider group.
- Practice where the effect of being repeatedly tested or observed can make participants behave or perform in a different way.
- Cohort effects in which participants may have certain characteristics arising from historical or cultural situations in which they have gone through and which may influence their behaviour and performance.
- Changes in the field of psychology, sociology or education, for example, new theories or modification of theories may affect the way research is perceived or interpreted.

1. ii) Cross sectional design

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This is a research in which groups of participants of different ages are studied at the same point in time. One can study children's academic aspirations by comparing at the same time responses of 5, 7 and 9 year-olds.

Strengths

- One can compare and understand behaviour of children of different ages in one go.
- It avoids problems of selective attrition, practice effects or changes in the field.

Problems of conducting cross sectional research

- One cannot study individual developmental trends or individual differences.
- If there is a wide age gap between groups being compared, cohort effects may occur.

GENERAL RESEARCH METHODS IN PSYCHOLOGY AND HUMAN LEARNING

Observation

1. Naturalistic observation involves recording behaviour of interest in the natural setting. This helps the researcher to observe things as they are or as they really happen in real life.

Strengths

- The investigator observes directly the everyday behaviour one wishes to explain.
- The investigator is able to collect other data related to the study as he/she carries out the observations.
- It is possible to get accurate data.
- Real, natural behavior can be observed and recorded.

Limitations

- One cannot control conditions under which participants are observed.
- Observer's bias may affect accuracy. Observer bias can include such things as people's attitudes or training which influence the way they perceive things.
- The presence of the observer may influence the way participants behave during observation.
- Observation does not tell us about cause and effect.
- It can be a slow and expensive method.

1. Structured observation involves observation of behavior in a laboratory setting where conditions are the same for all participants.

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Procedures used in systematic observation

- Specimen record which is a description of everything that is said and done during an observation session.
- Event sampling which involves recording all instances of a particular behavior during a specified period, for example every 15 minutes. One can, for instance, observe how many times a teacher praises children in a period of 15 minutes.
- Time sampling which involves recording whether certain behaviours occur during a sample of short intervals. One can have a checklist of the behaviours and the observation time is divided into a series of, say, 30 seconds or 1 minute intervals.

Strengths

- Each participant is granted opportunity to show the behaviour of interest without interference.
- It is possible to get accurate data.

Limitations

- May not yield observation of typical everyday behaviour.
- Observer's bias may limit accuracy.
- Observer influence may limit accuracy.

Self reports

1. **Clinical interview**

This is a flexible interview procedure in which the researcher obtains a comprehensive account of the participant's thoughts and behaviour by questioning the participant. Jean Piaget used this method in his studies. It is also used by doctors, psychologists, psychiatrists, and counsellors to collect information from clients.

Strengths

- This method comes close to the way participants think in everyday life.
- One can collect a lot of information.

Limitations

- The report could be inaccurate because of researcher bias or influence.
- It is difficult to compare individuals because questions posed to different people are not necessarily the same.

1. **Structured interviews, questionnaires and tests**

In these instruments, every participant responds to the same questions in the same way

Strengths

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Data collection and scoring is efficient.

- It is possible to compare the responses of the participants.

Limitations

- These methods do not always give the same depth of information as the clinical interview.
- There could be inaccurate reporting intentionally or unintentionally.

iii) Interview schedules

Interview schedules involve face-to-face and one-to-one sessions.

Advantages

- The interviewer and respondent are able to clarify issues that may be ambiguous.
- All questions are usually responded to.
- The interviewer is able to gather other data about the respondent and environment which may be useful in the research.
- Data collected is often more accurate than in questionnaires.

Disadvantages

- Interviews can be time consuming and expensive.
- Respondents may not be available when you want them.
- It is a slow method of data collection.
- Respondents may be intimidated by the presence of the interviewer and therefore not give objective responses.
- The interview might not be exactly the same from one respondent to another since new questions and clarifications may be needed from different individuals.
- Respondents might not give accurate information. It is not easy for the researcher to find out whether the respondents are telling the truth or not. Since the respondent is alone, her/his information is what the researcher takes as the truth.

Guidelines for doing a good interview

- Prepare the questions. Give them to another person to critique them.
- Make questions that are relevant to the study, clear, simple, short and as few as possible.
- Compile the questions in logical sequence.
- Study your questions, know them by heart to make questioning fast.
- Inform the respondents long before the day you intend to visit. Ask for feedback. Honour and respect their suggested dates.
- Be punctual, brief and to the point.

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- Build a rapport with the respondent before starting the real interview. You can do this by starting with introductions and explanation of the objectives of the study.
- Get a convenient, comfortable and quiet place for the interview.
- Be clear, polite, modest in dressing and when addressing the respondent.
- Avoid asking embarrassing and very personal questions.
- Do not push or force the respondent to answer a question. Respect his/her decision.
- Develop a quick way of recording. Have an assistant, take quick short notes or use a tape recorder if the interviewee agrees to it.

1. iv) Focus group discussions (FGDs)

Focus group discussions are similar to interviews but involve more than one person. They are discussions held with a group of people. The number should be 6-10. FGD is an effective method of data collection in qualitative studies, for example, those on opinions, attitudes, likes, dislikes or aspirations.

Strengths

- Plenty of data is collected within a short time from many people.
- The researcher and the respondents are able to clarify issues.
- The researcher is able to collect 'other' data related to the environment (human and physical) that may be useful to the study.

Most of the information collected is correct. This is because the group members tend to censure one another's responses. No respondent will dare tell a lie when she/he knows that those aroundher/him know everything about her/him.

Limitations

- The method is more time-consuming than individual interviews as more respondents are allowed to give their views on one question.
- Sometimes only a few people speak because some of the participants tend to dominate the sessions.
- The group may stifle or prevent a person wishing to give the true picture of the situation if this is not acceptable to the group.

Guidelines of doing a good FGD

- Make prior arrangements about the date, time, venue and the number and mix of respondents.
- Prepare the list of lead questions and make sure you are quite familiar with them.
- Get names of participants. Address them by name. Call out those who tend not to participate to ensure full participation.
- The group should be about 6-10 respondents.
- Have a group which has similar characteristics such as same sex, age, economic status, profession.

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- Have another person to record the responses. You can use a tape recorder but with clearance from the group.

v) Questionnaires

These are questions mailed or given to respondents to answer on their own. Questionnaires can be closed or open-ended. In open-ended questions, respondents are free to respond to questions in the way they choose. Respondents could, for example, be asked "What is the cause of discipline problems in our schools?" The answers are likely to vary from one respondent to another. In close-ended questionnaires, answers are provided and the respondent is asked to choose the answer. An example, "Indicate by an X the causes of discipline problems in own schools among the options given below." Answers to close-ended questions reflect facts and opinions more accurately. However, they give no room for other factors which could be significant and true.

An example of close-ended questionnaire/interview

1. Since completing your training as a teacher have you had any problems obtaining a job?
   Yes No
2. How often was this, the case?

Always Most of the time Hardly Never

3. What among the following fall within your immediate plans?
   Go to the university Look for another job
   Stay where I am Take leave for 2 years

An example of an open ended questionnaire

1. What would you like to do as soon as you complete the ECDE Diploma training?
2. What are the reasons for your choice of this action?
3. What else can you do if your first plan does not succeed?

Advantages of questionnaires

- Questionnaires are a quick method of collecting data.
- Plenty of data may be collected if mailed questionnaires are returned.
- Use of questionnaires is cheaper than interviews and FGDs.

Disadvantages

- It is not possible to clarify ambiguous questions in mailed questionnaires.
- In questionnaires, the researcher may not know how actually responded to the questionnaire.
- Respondents may give incorrect data if they do not understand some questions.
- Many questions may not be responded to arising in gaps in data.

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- Many questionnaires might not be returned.
- Questionnaires are mainly useful with literate people.
- The information received may not be correct because the person who responded to the questionnaire may not be the one who was expected to respond to it. The person to whom the questionnaire is sent may give it to another person to fill it.

Guidelines of developing a good questionnaire

- Questions should be relevant to the problem. They should be clear, short and simple.
- The questionnaire should start with a short explanation of the objectives of the study.
- Put questions in a logical sequence.
- Give another person the questionnaire to critique before compiling the final copy.
- Avoid questions that may require data that may be embarrassing to the respondent.
- Give the respondents time limit within which to mail questionnaires back to you.
- Enclose a self-addressed stamped envelop to quicken the mailing back of the questionnaire.

1. vi) Case studies and clinical method

The case study method is also referred to as clinical method in psychological studies. It involves getting a picture of an individual's psychological functioning. Data is obtained by combining interviews, observations, text scores and sometimes psycho-physiological assessments. The method is used a lot by doctors, psychologists, social workers and counselors. The case study method can also be used to study a group or youth group or an organization such as an ECD centre or primary school. Both qualitative and quantitative methods can be used to collect data.

Strengths

- It provides rich descriptive insights into factors that affect behaviour and development of an individual.
- It brings to light many factors that affect development.

Limitations

- May be biased by the researchers' theoretical preferences or experiences.
- The findings apply only to the individual who is studied.
- This method may be time consuming and expensive.

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- Sometimes data is not collected in a systematic way and therefore does not give a clear picture of the individual.

Psycho-physiological methods

Psycho-physiological methods measure the relationship between behaviour and physiological conditions. Some of the common methods take measures of autonomic nervous system (heart rate, respiration, and hormone stress levels) and measures of brain functioning such as electroencephalogram (EEG), event-related potentials (ERPs) and functional magnetic resonance imaging(fMRI).

Strengths

- They reveal which central nervous systems affect certain competencies.
- They help to identify the perceptions, emotions and thoughts of young children who cannot explain them verbally.

Limitations

- Many factors influence physiological reactions besides those that the researcher may wish to study.
- One cannot be quite certain how an individual processes stimuli or influencing factors.

Ethnography

In ethnography, the researcher becomes a participant observer of a culture or distinct social group. He lives with them during the research period. He participates in all their activities. He shares their life with them including their joys and sorrows. During his stay with them he collects extensive information on the culture of the people, their traditions, beliefs, values, aspirations and practices.

Strengths

- One gets a detailed description than can be obtained during a single observational visit, interview or questionnaire

Limitations

- Researcher's values and biases can influence the data obtained and how they are interpreted
- Usually the data cannot be generalized to other settings

Psychological texts

Psychological tests are used to measure and assess human behaviour. Some of the psychological tests are used to assess intelligence, vocational interests, values, attitudes, scholastic achievement or personality. A person's performance on these tests is compared to others in a similar situation. Some of the tests are administered on individual basis while others are administered in group settings.

Strengths

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- Many psychological tests are quite reliable as they consistently and repeatedly give accurate results.

Limitations

- Some of the tests are culturally biased since some of the items relate to experiences of a particular setting or culture.

Survey Method

The survey method is used to collect a lot of data from a large number of people within a very short period of time. Surveys are, for example, used to study types of diseases found in an area, opinions of people on discipline in schools, services available in homesteads, political opinions or attitudes. Surveys usually use questionnaires or interviews. Questions can be administered to participants by mail. Interviews are done through either telephones or person to person.

Strengths

- A sample when well selected, that is, it is representative of the population can be used as a basis for generalizing and predicting what is likely to be observed in future on the basis of current findings in the larger population.
- It is less expensive than using a study of the whole population.
- It can also be done within a relatively short time.

Limitations

- The results are likely to be inaccurate if the sample is not representative of the population.
- Sometimes returns of the questionnaires is poor and some questionnaires are not complete.
- People might not always tell the truth or give serious thought to their responses.
- Sometimes it is difficult to get in-depth information.

Experimental method

Details of this method are presented in Unit Two. Suffice to say that experimental research method is possibly the most scientific of the research methods. It tests for cause and effect of behaviour. It is crucial to understanding and predicting behaviour.

Strengths of the method

- It is precise.
- It is easy to replicate and verify.
- Comparison of the treatment and control groups makes the results more convincing.

Limitations

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- The method can be lengthy and rigorous.
- The method is best carried out by people trained in research.
- It can be expensive.

Self description

Individuals can provide important information for research studies through self description. They could write their life stories, through compositions, diaries, paintings and drawings or drama. The accuracy and reliability of such data will depend on how much the individuals want to disclose about themselves, the purpose and context of the research.

Correlation method

Ir. correlation method, two sets of variables or attributes are compared to see the extent to which they are related and if they can be used to predict each other. The correlation between two sets of variables is indicated by means of correlation coefficient which may be positive or negative. A positive coefficient ranges from 0 to 1.0 while a negative coefficient ranges from 0 to 1.0. One is a perfect correlation and 0 indicates that there is no correlation between two sets of variables. Details of this method are provided in Unit Two.

Achievement tests

Achievements tests measure what the student has learnt or the skills he/she has mastered. Teachers administer many achievements tests based on subjects during term time, end of term or year. External examinations conducted by Kenya National Examinations Council (KNEC) are also achievement tests. Achievement tests are described in more detail in Unit Ten.

Rating techniques

Rating technique is a type of observational or self descriptive method whereby besides categorizing behaviour or items to be observed, one also classifies behaviour according to evaluative characteristics. One makes qualitative judgments. Rating techniques use rating scales. The scale is a number of positions which are defined through brief descriptive statements.

Example of a rating scale

Indicate by an X your rating of the methods teachers use to maintain class discipline.

Method	**Often**	**Sometimes**	**Rarely**
Praises children			
Gives physical reward			
Pats children			
Scolds children			
Slaps children			
Beats children			
Allows children a lot of time to play			

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Strengths

- Can be done within a fairly short time
- It can give fairly accurate information

Challenges in use of rating scales

- Ensuring rater reliability. The rater(s) should have precise indicators of the behaviour or item to be rated so as to increase reliability. Reliability is better when the classification of the ratings is not too wide. A scale of 3-5 produces better results than a wider one.
- Rater bias. Some raters might be strict while others are lenient on the impression of the people or items being observed.
- Rating scales are not suitable for behaviours that do not occur frequently. The observation would have to be extended for a long time. This would be time consuming, expensive and would reduce reliability.

QUANTITATIVE AND QUALITATIVE RESEARCH METHODS

Definitions

Qualitative research collects information that shows how people in an area live and function as a community. The information collected consists of the views, opinions, feelings, values, norms aspirations, achievements and problems of a community group. Qualitative methods help to build an in-depth picture of how a relatively small group functions, how members relate and how different aspects of their lives are linked. Qualitative methods also indicate how people understand their own situation, their aspirations, problems and what their priorities are. Most of the data of qualitative research is presented through descriptions. The research describes what most of the people said or felt. It is difficult to use numerical form to analyze. Qualitative research does not use numerical data like percentages and correlation coefficients extensively.

Quantitative research is a method of carrying out an inquiry that emphasizes measurement. The data collected is analyzed using percentages, correlation coefficients and other statistical methods. This is because quantitative research methods collect data that can be analyzed in a numerical form. They pose the questions like who, what, when, where, how much, how often? Things are measured or counted or close-ended questions asked so that answers can be coded and analyzed numerically. Statistical analysis can be used on quantitative data to give precise description in terms of averages, ratios, ranges or percentages.

Characteristics of qualitative and quantitative research methods

Qualitative

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- The researcher needs to establish very good rapport to create trust so that people can share their intimate views, opinions and thoughts.
- The research poses 'how' and 'why' questions. It helps the researcher to understand social life of a community such as their opinions, relationships, traditions, practices and interactions.
- It is flexible. Usually questions which are asked are open-ended.
- Findings are usually analyzed as data is collected.
- The design of the study can be modified as the research progresses to follow up significant findings as they emerge.
- Qualitative methods are used in flexible way but should be systematized to ensure that the findings are objective and reliable.

Quantitative research

- It is used to collect data that is counted or measured.
- Data is analysed using statistics.
- Design and methods are more systematic and standardized throughout the research.
- The set design usually holds for all the aspects of a particular research such as data collection and analysis.
- Quantitative research places emphasis on methodology, procedure and statistical measures to test hypothesis and make predictions.

Strengths and limitation of qualitative and quantitative research

Qualitative research

Strengths and usefulness

- When planning a programme concerning social change.
- To get an in-depth understanding of a particular social context. Information collected consists of people's views, opinions, values, norms, practices and aspirations.
- To get people's perceptions and opinions.
- To get relevant indicators of qualitative change according to the target group.
- When time and money are limited.
- When addressing sensitive and personal issues.
- It is easy for unskilled researchers such as community members to participate in this type of research.

Limitations

- A researcher can be derailed to collecting a lot of unnecessary data.
- Data are not as precise and objective as in quantitative research.

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Quantitative research

Strengths and usefulness

- When accurate and precise data are required.
- To get a broad view of a population.
- To identify major differences in characteristics of a population.
- To find out if there is and the extent of a relationship between problems and causes.
- Provide hard (statistical) data to prove that a problem exists or to show results of an intervention.
- Establish clear baseline data that can be used for evaluating impact of an intervention.
- Data tends to be more convincing to especially policy makers because real numbers and comparisons are presented.

Limitations

- Can be expensive
- Require skilled researchers and data analysts.
- May ignore relevant views of respondents, when closed ended questions are used.

Activity

- Define qualitative and quantitative research
- Compare and contrast qualitative and quantitative research in terms of

Characteristics

Strengths

Limitations

Qualitative research methods

Qualitative research uses a variety of methods. Some of these methods including observation, interviews, personal description, focus group discussions and case studies are described in Unit Three. This unit will present additional methods used in qualitative research. These are the main methods used in the Participatory Learning and Action (PLA) research. PLA is discussed in greater details in chapter 5. The PLA qualitative research methods include mapping, transects, historical profiles, seasonal calendar, daily calendar, ranking and scoring, Venn diagrams and study of secondary information.

Social and resource mapping

Social and resource mapping consists of the community members drawing a map to show the location of physical features (mountains, rivers, valleys) and resources are available. The resources they map include homesteads, water

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sources, institutions (schools, health centres), transport network, shops and factories. A social resource map enables community members to analyse and understand their area better. They are able to appreciate what they have. They also reflect on how they can use what they have better. A social map could indicate differences between what families possess; differences among community members in such aspects as educational level, ethnic groups, religious beliefs, traditions and property owned.

Procedure

- A community group chooses a suitable place for the mapping and agrees on the boundaries of the area. The map can be done on paper or on the ground if the participants are not literate.
- Local materials can be used as symbols to represent important features such as roads, schools, churches, shops, markets, rivers or wells, hills or homesteads.
- Encourage everyone to participate in the mapping exercise.
- A literate participant should copy the map drawn on the ground on paper.
- Discuss the feelings and views of the members about the exercise.

Advantages of mapping

- Encourages participation by many people.
- People discover their resources. They understand where various features are situated and relative distances between different features. They reflect on how they can utilize what exists better.
- Encourages innovation and creativity in use of local resources.
- People gain more information and understanding of their community.
- The map can lay the foundation for more analysis.
- It stimulates discussion about resources, opportunities, challenges, problems or livelihood.

Activity

- Draw a community map of your village. Using symbols, show all the resources and opportunities that can enhance learning and development of young children.

Transect walk

A transect is a walk across a section of the area occupied by a community. The group that makes the transect walks observes and learns about the physical features, soils, vegetation, institutions and services, infrastructure (roads, telephone lines, electricity lines) and land use patterns.

Procedure

- The group decides the direction and distances of the area to be covered which are likely to give greatest variety of information about the area.

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- The group decides what should be observed such as vegetation, buildings and land use, number of children, play space.
- The group leaders discuss the criteria to be used to divide the area into zones, for example, vegetation, and density of households or land use.
- The group leaders assign various sub-groups or individuals to observe certain features during the walk.
- The group should walk slowly, look carefully, and discuss what they observe and record it.
- After the walk, a transect diagram indicating the main features noted during the walk is drawn.

Advantages of a transect

- Allows more detailed information to be collected than during the mapping exercise.
- Facilitates analysis of what is available and how it is used.
- Allows the community to appreciate the resources they have and differences within the area.
- The transect diagram provides a visual impression of the area.

Historical profile or timeline

A historical profile shows the main events that have occurred in a community within a certain period of time. It shows trends, patterns and changes over time, recurring and one time events. Changes can be cultural, social, economic or geographical.

Procedure

- Form groups according to their interests or age.
- Use focus group discussions. Older members of the community can describe changes which have occurred in their community since they were young in such aspects as vegetation, organization of the community, child rearing methods, buildings.
- Use major events such as year of independence, great famine or floods as the point of reference rather than actual dates. People are more likely to remember major events and what happened at the time.
- The group should discuss and record the historic events which are important to them. The group could discuss the problems related to the issues under consideration, causes of the problems and how people dealt with the problems. It could also discuss how the events affected people's lives, lessons the group can draw from the events and how that knowledge can help planning for the future.
- Let the group decide if it wants to use symbols and visuals in their profile.

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- Additional information can be sought from reports, books and media to get more data about the events to be recorded in the historical profile.
-

A A historical profile of Kenya 1991-2006

Year	Event	Effect	Coping mechanism
1991	Ethnic clashes, floods	Displacement, death, suspicion	Resettlement, relief supplies, reconciliation
1992	General elections	Suspicions	Campaign, media
1997	Ethnic clashes/ general elections	Displacement, suspicion, drop in tourists	Resettlement, reconciliation campaign, advertisement for tourists
2002	General elections	People hopeful	Rehabilitation of cash crops/ livestock and industries
2005/6	Drought	Lack of food, inadequate water, increase in childhood illnesses	Emergency food, relief by government, NGOs and bilateral bodies

Advantages of historical profiles

- The community can understand and appreciate major events and changes that have occurred within a time span.
- The community reflects on how these major events have influenced their lives.
- The community can try and identify the lessons they learnt from both positive and negative major events. This information can be used to avoid problems in the future.
- The community can relate the effects of certain events on the present and possibly future conditions.
- Facilitates dialogue and critical analysis of issues.
- Provides information that can be used to plan future activities.

Seasonal calendar

Aseasonal calendar can be used to analyze and record different conditions and activities occurringin different seasons. Symbols can be used to represent months or seasons and the activities occurring in different seasons. The group can start by recording the major features of seasons, for example, rain or temperatures. Then they can record the other aspects of interest.

Month	***Weather conditions***	***Diseases & conditions***	***Activities and conditions***
January	Hot, dry	–	Harvesting, enough food, enough pasture

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February	Hot, dry	–	Enough food, lack of pasture
March	Hot, dry	–	Preparing land, lack of pasture
April	Warm, rain	Malaria	Planting, enough pasture
May	Warm, rain	Inadequate diet	Weeding, enough pasture,
June	Cool, dry	Colds and pneumonia	Weeding
July	Cool, dry		Weeding
August	Cool, dry	Enough food	Harvesting, enough pasture
September	Hot, dry	Enough food	Preparing land, lack of pasture
October	Hot, dry	Inadequate diet	Planting, lack of pasture
November	Cool, rain	Malaria	Weeding, enough pasture
December	Cool, rain		Weeding, enough pasture

Advantages

- It helps to compare community activities.
- It can be used to analyze conditions that change with seasons, for example, incidence of disease, pests, lack of food, shortage of water or pasture.
- It can be used to establish relationships and causes of events or happenings. For example, the community may be able to find out that heavy rains are followed by an outbreak of malaria due to breeding of mosquitoes in stagnant water or diarrhea diseases due to people drinking dirty water from polluted rivers. They can also establish causes, for example they may be able to find *j* out that cutting of trees results in less rainfall and more soil erosion in the area.
- Opportunities and resources available in different seasons can be identified.

Daily schedule

It is used to identify who does what during different times of the day. It shows the number of hours people in a community spend doing different activities like working and relaxing. It also shows the division of labour by gender. It can be used to create awareness on the fair sharing of responsibilities between gender. Different symbols can be used to represent different activities and time.

Daily schedule for men and women

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R**anking and scoring**

Ranking means putting various issues of community life in order of importance, value or preference. Scoring means assigning values to different issues according to some convenient scale. In participatory community development, people are often asked to rank priorities, problems, needs, constraints, preferences, or resources.

Procedure

- Ask the group to list problems or resources.

Participants identified large families as the biggest problem followed by poverty.

- The group does the ranking using preference, pair wise or matrix scoring procedures.
- In preference ranking, group members can discuss and agree on the ranking from the most to the least important. If group members are not literate, they could use objects to indicate their preferences. The most prevalent or important factor would have the most number of objects reducing according to reduced preference up to the least prevalent or the one judged to be of least importance.

Preference ranking done on individual basis

Problems	**P_1**	**P_2**	**P_3**	**P_4**	**P_5**	**Total**	**Rank**

P: Participant

Participants identified large families as the biggest problems followed by poverty.

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Diseases	1	2	1	4	5	13	3
Poverty	3	3	2	3	3	14	2
Access to education	5	1	3	2	1	12	4
Sanitation	1	4	4	1	2	12	4
Large Families	4	5	5	5	4	23	1

Pair wise ranking

This is done by building a matrix where the problems, for example, are listed on the vertical axis and on the horizontal axis as shown below. Pair wise ranking gives better results when done on the individual or small groups and comparing a few problems at a time.

Pair wise Ranking

	Disease	***Poverty***	***Education***	***Sanitation***	***Large Families***	***Poor Organization***
Disease	X	Disease	Disease	Disease	Disease	Disease
Poverty	X	X	Poverty	Poverty	Poverty	Poverty
Education	X	X	X	Education	Education	Education
Sanitation	X	X	X	X	Sanitation	Sanitation
Large Families	X	X	X	X	X	Poor Organization
Poor Organization	X	X	x	X	x	x

Procedure

- Ask the group to answer the question" which problem is more important for you and why? "They answer by comparing two problems by taking the first one listed on the left with the second one listed on the top, for example, disease on the left (vertical) with poverty on the top (horizontal). Continue across the row.
- The specific answer is recorded in the cell. If, for example, the group compares disease and poverty and regard disease to be the more important factor. Then record disease in the cell.
- Also record elsewhere the reason why factor was prioritized.
- Complete the comparison for all the factors.
- Add up the number of times a problem was identified as more important than the other problem.
- Prepare a matrix to summarize the number of times each factor is prioritized and its rank.

Problems ranking based on pairwise ranking

Problems	Score	Rank
Disease	5	1
Poverty	4	2

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Education	3	3
Sanitation	2	4
Large Families	0	6
Poor Organization	1	5

Advantages of preference and pairwise ranking

- Preference ranking is a quick method of establishing people's views about issues.
- It can be used in prioritizing issues by comparing views of different individuals and groups
- The groups will most likely abide by the findings since each is given an opportunity to make a choice.
- In pairwise ranking, more authentic priorities are established since two factors are compared at a time.
- People have an opportunity to discuss, analyze and prioritize needs, problems and problem solutions.

Matrix scoring

In matrix scoring, problems, constraints or solutions are compared against selected criteria.

Procedure

- Develop a matrix by recording factors or problems along one axis. Motivate the group to find criteria according to which they can differentiate problems. For example:
- Causes of the problems: What is so bad about this problem? Possible solutions: Is there a realistic solution to this problem?
- List the criteria developed by the group on the other axis of the matrix.
- List all criteria either in positive or negative form
- Choose an appropriate method of ranking. For instance in the example below, rank each candidate in order, 1 being the most preferable and 3 the least preferable according to each criterion. Get the overall ranking by adding the numbers in the three columns and calculate.
- Yusuf is the best candidate according to the ranking, that is, he had the least mean score.
- The group should discuss the results of their ranking
- Ask them to choose the one problem or solution they think is most important and why.

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Ranking candidates who have applied for a job

Creteria/ Ranking	Alima	Yusuf	Adan
Hardworking	2nd	1st	3rd
Intelligent	1st	2nd	3rd
Team Player	3rd	1st	2nd
Total	6	4	8
Average	2	1.3	2.6

Advantages

- Data from the ranking and scoring is numerical and therefore can be quite convincing.
- Encourages intensive discussion on the importance of problems and criteria for choices.
- Reasons for local preferences are understood since people develop criteria used in making choices for the ranking of problems.

Venn or chapati diagram

Venn diagram shows relationship between actors, institutions or factors. The relationship is indicated by the position of circular shapes in relation to each other. The importance of the ft or institution is shown by the size of the circle as perceived by the key stakeholders.

Procedure

- The group identifies the organizations or institutions which are important to the community which are working there.
- The group should draw a big circle to represent the community or institution. Then put smaller circles inside the large circle to represent-the-organizations/institutions inside the community or those involved in the institution. Put other circles outside the community/institution to represent those organizations /individuals outside the community/institution who influence community/institution. The small circles represent the institutions or organizations which have little importance and impact. The larger circles represent important and effective institution organizations.
- Distance between the circles indicates degree of cooperation or contact.

Interpretation of circles by size or position

Separate Circles	Little or no contact
Touching circles	Information passes between institutions
Small overlap	Some cooperation in decision making. Resources and implementation

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Large overlap	A lot cooperation in decision making, resources and implementation.

- After drawing the diagram, discuss and analyze the results to find out which institutions o individuals have great or little influence and why, how they can be more involved and their linkages be strengthened.

Creating village resources management plans (VRMPs)

Purpose

The plan serves a variety of purposes and audiences.

1. It is a record of community's development projects. These projects are ranked in order of their importance from the first to the last. The plans also show the resources available in the community which can be used to carry out the projects identified. This record is used to plan activities to be carried out in the development projects.

1. The Sub-location Development Committees use the document for planning and implementation of community projects. They forward it to the District Development Committee for possible funding.

- The plan can help external donors and implementing agencies to choose the projects they would want assist the community to implement. The projects chosen by different agencies are those in line with their goals.

The plans contain several elements:

1. Development priorities as agreed on by the community.
2. Proposed activities and resources required to implement these activities.
- Duties and responsibilities for individuals and groups.
1. Work schedules.
2. Identification of areas where the community needs external assistance.
3. Costs of various activities.

Study of Secondary Information

Secondary data is information that has already been collected and documented by others. The main sources of secondary data are (i) published sources and (ii) unpublished sources

Published sources

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- Official government publications such as ECD policy framework, Children's Act, syllabi, reports of Education Commissions and Taskforces, and conference reports.
- Publications from universities and other research institutions.
- Reports from development agencies, government departments, NGOs and schools.
- Newspapers and periodicals.
- Books in libraries and other institutions.

Unpublished sources

- Reports and documents from government departments, NGOs, private sector and institutions.
- Researches in universities and other institutions.

Strengths of the method

- One can describe and explain change by analyzing data collected at different times on similar issues.
- One can get a rich source of information to compare different environments.
- One can save time and money by using secondary data.
- Use of secondary information helps to avoid duplicating previous studies.
- Information helps those implementing projects to avoid making the same mistakes made by others in the past.

Limitations

- Secondary data is not always as accurate as field data
- The original investigator might not release all the data available.
- The original investigator may have had a specific goal for the research and this could introduce a bias in the data.
- The researcher may have insufficient information about how data was collected. This information is important for determining potential sources of bias or errors.

Creative arts

Data can also be collected through creative arts such as drawings, use of symbols, songs and symbols, songs and drama these methods are particularly useful when collecting data from children. drawings and symbols can be used to illustrate something one is trying to explain, illustrate ideas or criteria for

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preference. these drawings and symbols can reveal ones feelings, aspirations or life experience. in songs and drama, participants can reveal their emotions and feeling, concerns, priorities, level of awareness and life experiences.

Quantitative research methods

Quantitative research methods include structure interviews questionnaires, rating scales

,observation, psychological and achievement tests. These methods are discussed in unit three and unit ten study of secondary information can also provide quantitative data

PARTICIPATORY LEARNING AND ACTION.

Definition of participatory learning and action (pla)

Pla is both a philosophy and also an approach of bringing about change and development in communities. it is a philosophy because it is based on the principle that community members can brings about their own change and development. People are able to analyses their needs, aspirations, goals, problems and resources. they are able to come up with solutions to solve their problems. they are also able to identify what to do to meet their needs.pla is an approach because it uses different methods to collect data. these methods include for example

Interviews, focus group discussions (fgds),social mapping, seasonal calendars,venn diagrams and transect walk. It is a form of research which considers the interrelatedness of different aspects of life of a community for example the relationship ,economic activities disease and skills.

ORIGIN AND DEVELOPMENT OF PLA

PLA has been adapted from participatory rapid assessment (PRA) research. PLA is a type of qualitative research .the term participatory is used in this research because the community members participate in the research. They participate in analyzing their own situation. This involves identifying their needs problems, aspirations and resources. They also identify they can carry out to meet their needs and what they can do to solve their problems. They participate in planning and implementing activities to solve the problems they identify in the research. The term”rapid”was used because data was collected in a relatively shorter time than is the case with ordinary researches.the”assessment”is used because this research helps all those involved in it(both outsiders and the community) to gain deeper understanding of the community’s way of life, for example,culture,traditions,practices,aspirations and feelings PLA has however continued to be improved and refined because of new ideas and theories.

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The term PLA is now preferred. This is because community members participate in the research and all their development projects. As they participate they also learn. They do research and also decide on the actions or activities to carry out to meet their needs and to solve their problems. Hence the term PLA. It comprises the methods used in rural and urban research. In addition, it stresses the importance of community learning from the research and taking action. It is not enough to collect data. The community should reflect on the data, learn from it and take action. The action taken by the community involves planning and implementing activities which improve the community's social and economic conditions. In recent years, there have been changes in the way participatory processes are used both in the community as follows.

Participatory processes are used both in the community as well as in national and international decision-making processes.

- The process is user in program development and not just in projects.
- PLA recognizes and respects differences in the community and power structures that exist.
- There is more recognition of the need to assess quality and impact of participation rather that just promotion participation.

Why use the Approaches?

- To ensure that the data collected , the way it is analyzed , the conclusions reached and the recommendations made are owned by the community.
- To provide opportunities for communities to learn. They acquire new knowledge and skills as they participate in the research.
- To ensure that the communities are involved in decision-making on issues that affect them and also in their own development.
- To make the communities accountable for their own development. They take responsibility of the mistakes they make. They also celebrate their achievements.

Importance of PLA approaches

- Participatory learning and action (PLA) approaches are user- friendly.
- They encourage community members and outsiders to work together and to share ideas and experience.
- PLA enables the community members to share their secrets and sensitive issues without feeling embarrassed
- The process arouses people's interest and keeps them motivated throughout the exercise.

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- PLA tools enable community members to explore and reflect on their own lives, their culture and traditions in a creative way.
- PLA processes help to build friendships between the community members and outsiders.
- PLA gives community members the power to influence development activities that affects them.
- PLA helps the communities to understand and appreciate themselves better. They are able to identify the resources such as skills, knowledge, money, time, facilities and other strengths which they have and which they can use to meet their needs and solve their problems. They are also able to identify the challenges they face when implementing development projects.
- It ensures that community members understand the project objectives and activities, and therefore are more likely to be committed to the project.
- PLA helps the outsiders to understand better the way the communities think, their attitudes and aspirations.
- It makes it possible to collect information on various aspects of community life.
- The findings are presented in such a way that makes it easy for the community to understand.
- Results are often obtained quickly.
- Relevant indicators showing qualitative change are identified.
- As people participate in the research they are able to identify emerging issues and address them.
- People learn as they undertake the research.
- People feel less threatened by PLA methods than by formal research methods. This is because they are allowed to participate, make decisions, ask questions and make suggestions.
- Research is part of the development activities in the communities. The findings are used in planning and implementing community development activities.
- It enables the powerless and voiceless like women, youth and children to participate, to speak out and share their ideas and experiences.

Characteristics and principles of PLA

1. PLA is a systematic learning process. The community members and "outsiders" learn as they participate in the research. They acquire new knowledge and skills as they carry out research and as they share ideas and experiences.

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2. PLA recognizes that there are different ways of interpreting issues and solving problems. All people must therefore be listened to. They should be given the freedom to contribute their ideas, analyze their problems, identify possible solutions and decide which solution is the best for their problem.
3. Multidisciplinary team. The PLA team has people with different skills and backgrounds. This ensures that they deal with different issues in an integrated way. This strengthens the PLA activity. All the members of PLA should be involved in all the stages of the study. Community members should be in the PLA team.

Multidisciplinary teams
Multiple disciplines

Men/Women Insiders/outsiders

4. Triangulation in PLA means the same data is collected from different sources using different methods and tools. The different sources include parents, teachers, community leaders and religious leaders. Examples of different methods include FGDs, daily schedules, transect walk, seasonal calendars, Venn diagrams and community maps. Examples of tools include interviews, FGDs, participant observation and self reports. Triangulation ensures the authenticity and accuracy of data. The use of different sources, methods and tools to collect the same data ensures that data collected is cross-checked and verified./ this ensures that data is valid, accurate, genuine and convincing.
5. PLA requires flexibility. The study needs to be continuously reviewed so that methods are adapted to the changing needs of the community.
6. PLA is a group learning process. PLA must involve the community members and "outsiders" at all stages. This helps in better interpretation of data and understanding of issues.
7. PLA should be context specific. This means that the methods and approaches are designed or adapted to the local situation, preferably by the community members with the assistance of outsiders, this ensures that the methods and approaches are relevant, more accurate and owned by the people.

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8. The "outsiders" should act as catalysts. They motivate the community to participate in the research. They ask questions which make the community members start to reflect on their lives, traditions, culture and practices. The community members should decide what to do with the data and information that is collected. Both the "outsiders" and the community members should be committed to engage in follow-up actions.
9. Make the study simple. Data collection should be simple, and time spent on data collection should be minimized. Only essential data should be collected. Data analysis and reports should be simple and easily understood by the community.
10. On the spot analysis. The data should be analyzed as it is collected. This provides opportunities for community members to learn as they participate in the research. This is also allows emerging issues to be identified early and action taken to address them.
11. Offsetting biases and being self-critical. PLA team needs to ensure that the study caters for all groups in the community without biases.

The three pillars of PLA

PLA is supported by three pillars (i) Behaviour /attitudes (ii) Methods and (iii) Sharing.

Behaviour/ Attitudes

Pillars of PLA

Sharing

Behaviour / Attitudes

PLA challenges both the experts and that community members to change their attitudes and behavior towards one another and towards life issues. It does this by ensuring that those participating:

- Stop believing in stereotypes of the society such as, gender related stereotypes. For example, the belief that women do not think, they are ignorant and meant to be shy. Another stereotype is that children "should be seen and not heard." Women and children should be involved in the decision making processes on issues that affect them. In addition, educated people believe that people who are illiterate are ignorant and know nothing. They also believe that illiterate people should not participate in researches and development activities.
- Embrace Error. It is important to accept that people will make mistakes as they participate in the research. When they make mistakes let them know they have made mistakes. Help them to learn from these mistakes.

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- Listen to people actively. Listen to their views, what they say in words and emotions they express. Respect their views, ideas, emotions and culture. Help them to analyze their views, ideas, and culture so that they identify what is positive and what is not positive. Avoid criticizing them.
- Accept people as they are and respect their individuality and differences.
- Hand over the stick. Learn to hand over responsibilities. Allow people to take charge of their own affairs. Allow people to control their destiny. Let the old people, for example, accept that they can handsome their responsibilities to the youth. Those in position of responsibility should also accept to hand over power to others.
- Know you are there for them not for you. Take time with them.
- Do not underestimate people's ability. Appreciate that people are the greatest resource that you have.
- Use your judgment wisely at all times.
- Be flexible and move with the pace and style of the community. Be patient with their pace.

Methods

PLA uses methods and techniques that allow those participating to interact and to share their knowledge, skills and experiences freely without fear and intimidation. These methods emphasize:

- Dialoguing on ideas not personalities.
- Use of mapping and diagrams for collecting data and not interviews alone.
- The use of ranking, comparing and scoring to analyze data.
- Presenting to the communities data collected in order to make them feel they are valued and they are not being manipulated.
- Working with the community to prepare their community Action plan and implementation it.
- Monitoring and evaluation of all activities carried out in order to find out whether the objectives of the program have been achieved or not.

Sharing

PLA encourages and emphasizes:

- Mutual sharing of ideas, knowledge, skills and experiences between community members and outsiders without fear.
- Sharing knowledge between community member and outsiders.
- Analyses of data by community members and outsiders
- Sharing and offering hospitality by community members.

The steps in PLA process

1. Site selection. The area where PLA is to be carried should be:

- Accessible to all
- Agreed upon by all

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2. Selection of PLA team including community members. The PLA team should

- Be gender balanced
- Cater for different groups in the community that matter, for example, representatives ;«-youth, religious bodies, children, women and people with disabilities.
- Have people with different skills, experiences and backgrounds
- Have people who are willing to be trained and to participate in the research.

3. Preliminary visit. The PLA team should make a preliminary visit to the research site to establish contacts, know the people, meet community leaders and sort out administrative issues.
4. Organize a public meeting to officially launch the PLA activity with the government: and other key leaders. During this launching explain objectives, the site and activities of PLA
5. Data gathering with the people. Before beginning to collect data, clarify objectives, choose main topics or themes, identify indicators and sources of data, identify and develop tools. The s-t activities can be carried out in a workshop. The community should be well-represented. This involves collection of data from the field and from records. During data collection ensure:

- Respect of everyone's contributions.
- Punctuality for all activities. Adhere to times given.
- Observe and respect community work schedules, traditions and culture.
- Make plans that suit community activities and lifestyle.

6. Data analysis with the people

- Identify and list problems, their causes and opportunities.
- Analyze the problems being encountered by the community
- Rank the problems.
- Identify the opportunities available to deal with the problems prioritized. The opportunities include skills available in the community, money, time and facilities.

7. Preparation of Community Action Plan (CAP) with the community. The PLA members and community groups discuss further the problems, opportunities and interventions identified. A Community Action Plan is then prepared.
8. Adoption of the CAP. The community discusses in details the CAP, activities to the carried out, roles of different members, resources required and other implications of the CAP. The community identifies the available resources and those required from outside the community. The

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communities must understand clearly their role and that of the external donors.

9. Implementation of the CAP. The community should decide on the implementation process. Members should understand clearly the activities to be carried out, when and by whom. The structure to oversee the implementation should be agreed upon.
10. Monitoring and evaluation. The PLA team should train the community in basic elements of monitoring and evaluation so that community members too monitor and evaluate the PLA activities. The community needs to develop its own monitoring and evaluation indicators.

Collecting data from the field

- Find out how people begin activities.
- Let local people lead the process.
- Avoid bias. Include all categories of people, both rich and poor (all social classes), young and old , men and women.
- Use triangulation (cross -checking) method. Collect the same data from different sources for example parents, teachers, youth, community leaders and use different methods/tools.
- Allow information to unfold using insights gained to direct and redirect research.
- Look for diversity, exceptions, contradictions, patterns and variations to patterns.
- Recognize the principle of optional ignorance. This means that you should accept you do not have to know everything to know something.
- Use a wide range of techniques to get enough understanding of the community.
- Local people should do the research. The outsider should be a facilitator. This helps to build local expertise which will result in acquisition of new knowledge and skills by those participating. This helps to develop sustainably.

Activity

You have been asked by your community to help them establish an ECDE centre.

- Explain the steps you would follow to ensure that the community is fully involved in the process.
- Why would the involvement of the community be important?

PLA techniques and tools

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PLA tools and techniques are user-friendly. They also try to eliminate biases such as:

- Meeting only certain people and leaving out others.
- Coming up with pre-conceived biases and prejudices
- Being in too much hurry. Community involvement requires time and patience.
- Disregarding cultural and traditional values
- Misinterpretation of people's lives, their ideas, beliefs, practices and culture through translation.

The following is a summary of different tools which are used to collect data in PLA researchers (These tools are described in detail in Units Three and Four)

Data collection tools

1. Spatial tools which give us information about the areas of study

- Community maps – social resource map
- Transects
- Household sketch map
- Mobility maps

1. Time related or temporal tools which are used to gather information related to changes over time

- Timelines or historical Profiles
- Trend lines
- Seasonal Calenders

1. Socio-economic tools which are used to collect data on livelihood and ways of life.

- CailyCalender
- Venn/Chapati diagrams
- Livelihood diagrams
- Focus group discussions
- Semi structure d interviews
- Indigenous technical knowledge

Analysis Tools

- Pairwise ranking
- Preference ranking
- Direct matrix ranking
- Flow diagrams.

Guidelines for analysis of PLA findings

- Data should be reviewed continuously as it is collected. It is classified and analysed. This to identify emerging issues. More questions are developed to follow up these issues.

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- Prepare list of key issues and questions to help sort out the findings, identify patterns, differences, variations and contradictions.
- Formulate questions based on issues and try to answer them using the data.
- Discuss each topic or issue, summarize results and draw conclusions.
- Use diagrams, matrices and ranking and other analysis tools.
- Tabulate information for further clarification. Discuss any issues that you think are imp with the community and the experts.
- Be self critical. Assess yourself to find out how well you are managing the activities and the community.
- Findings should be consistent and not contradict one another. If the findings contradict secondary data or findings from other researches you should try to find out why this is so.

Completing the process

- Appropriate action plans should be drawn.
- Identify procedures for carrying out the activities.
- Define roles of different players.
- Define monitoring indicators and procedures.
- Identify resources such as skills, money, facilities to be used in the project. Suggest ways of mobilizing them.
- Have a written plan. A suggested format: objectives, activities, roles, resources, finances,

Time-frame.

Summary of benefits of people's participation

- Ownership of the development activities. People become more interested and committed activities being carried out when they are involved in making decisions. They are willing to change the way they work or behave if they are involved in designing these changes.
- The development activities are relevant to the needs of the communities because they address the needs and problems identified by the community members themselves.
- All groups in the community including children, women, the poor and vulnerable involved in the development activities.
- The community will commit time, emotions, resources, skills and knowledge to ensure the success of the programme. They will also acquire knowledge and skills which help them to continue with PLA and other development activities on their own.

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Challenges and dilemmas of PLA

- The results only apply to the community involved in the project. It is not easy to generalize them to other situations.
- Participants of the research can be biased if all the groups are not well represented and g opportunities to express their views.
- If not systematically organized, the results are likely to be impressionistic with no backing of statistics.
- Results of qualitative studies might not carry as much weight as quantitative studies with decisions makers. This is because most decision makers prefer to base their decisions on statistics since these are easier to prove.
- Professional bias versus multisectoral and interdisciplinary emphasis in the PLA approach.
- There is need to remove divisions across professionals and start integrating and linking up with other professionals from other fields of specialization.
- It takes time for changes to take place in the community. This is because PLA is a very slow process. One needs to be very patient before positive results are noticed. PLA tends to raise high expectations among the people. People expect high and quick rewards for the time they spend in the PLA process. Such people may abandon the project if their expectations are not met quickly.
- The PLA team needs to have members who have a wide experience in group dynamics and adult education methods. Getting such qualified people who are willing to work at the community level can be difficult and expensive.
- PLA requires facilitators who have a lot of patience. They need to look, listen and learn. This takes time and challenges the patience of facilitators. It is easier and faster to tell people what to do but this does not bring about community development.

Activity

A new youth group in your community wishes to start an environmental conservation project for income generation. The group has asked you to advise it how to go about the project.

- Explain to the group why it should use PLA approaches.
- Explain to the group the steps to use to start the project.
- Point out some challenges the group should anticipate.

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Appreciative inquiry

Appreciative inquiry (AI) is an approach which emphasises the communities focusing on their abilities, potentials, strengths, achievements and resources rather than on their problems. AI goes beyond participation of the community. It encourages the development of self-esteem and confidence at the grass-roots level. Appreciative approach came into being with the understanding of the importance of people's self reflection. It encourages us to appreciate what this world can do for us and what we can do for it. AI encourages the communities to find out what exists (what is) and that they are proud of. The search of what exists helps the community members to realize that they have many resources which they can use to bring about their own development. These resources include knowledge, skills, abilities, potentials, time and facilities. AI also encourages the community members to have dreams and aspirations (what could be) that motivate them towards working to improve their future. AI therefore acts as a catalyst or a motivating force for community development. What communities have makes them feel proud of who they are while their dreams and aspirations give them hope for the future. These energize or motivate them to take action in order to improve their present and their future.

When communities realize that they have many resources and strengths they feel energized and motivated. On the other hand, if they focus on problems they tend to feel discouraged.

Stages of appreciative inquiry

The appreciative inquiry involves the community working together to collect information on their strengths, abilities, good news, success stories and achievements. The communities also spend time to appreciate themselves and celebrate all their successes and achievements. This makes them feel good about themselves. AI also involves our ability to recognize the strengths and achievements of systems, situations and other human beings around us. Local people can use their understanding of "the best of what is" which includes their strengths, potentials and resources to develop a vision of what their community might be in the future. They achieve this goal by using their current achievements as a foundation to develop their dreams and aspirations to improve their future.

Discovery

"What gives life?

(The best of what is)

APPRECIATING

The appreciative inquiry cycle

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Discovery phase

The main aim of the discovery phase is to help the community to appreciate the best of "what is" by looking for best examples of their victories, strengths, achievements and successes. The community then tries to understand what made it possible for them to achieve the victories, successes achievements identified. Was it leadership, internal relationships, knowledge, skills, technologies values, capacity building or external relationships?

People deliberately choose not to focus on problems and challenges. They do not spend timeanalysing problems and deficits. Instead they focus on strengths and achievements, systematically seek to isolate and learn from even the smallest victories and successes. people share stories of exceptional achievements and they discuss the factors which made realize these achievements. They deliberate upon the aspects of their history that they value and want to enhance in the future.

Dream phase

This phase involves the community sharing their dreams and aspirations. They base these dreams and aspirations on their past victories, success and achievements. They share their experience their past achievements, victories and successes. They believe firmly that if they have succeeded the past they can achieve greater victories and success in the future. This belief becomes a d force which helps them to march into the future with confidence that they will succeed. They usethese positive stories in the same way an artist uses paints to create a portrait of the community potential. They think great thoughts and create great possibilities for their communities, then turn those thoughts into dreams and aspirations for themselves.

Design phase

The community develops ways of ensuring that they achieve their dreams and aspirations. They do so by re-defining approaches to leadership, governance, use of resources, and participation on capacity building. They identify qualities of community life that they want to protect, and the relationship that they want to achieve. They also identify the activities they need to carry out in order to ac their dreams and aspirations.

Destiny phase

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This involves the community working together in order to ensure they continue to do all that
required to achieve their dreams and aspirations. It is a time of continuous learning, making adjustments and doing all that must be done to ensure dreams and aspirations are achieved. During ibis time people have a lot of motivation and energy for being innovative and creative in order to achieve the goals they have set for themselves. The destiny phase leads naturally to new discoveries of community strengths, new achievements and beginning of the whole process a new.

Why AI works

- It integrates different ways of knowing.
- It allows room for emotional response as well as intellectual analysis, room for imagination, innovations, creativity and rational thought.
- It raises people's self-esteem and confidence.
- It makes people proud of what they have and who they are.
- It gives positive feelings about people and their environment.
- It raises hopes for better future.

Principles of AI

The following principles help explain the power behind the appreciative inquiry approach

- *The constructionist principle*suggests that social knowledge and community destiny are interwoven. It is important to know people's achievements and best times as well as their hopes and dreams. If you stand on the foundation of achievements you are more likely to move on into higher levels of achievement and success. We co-create reality through our language, thoughts, images and beliefs and also when we ask questions and when we intervene. "The seeds of change are implicit in the very first question we ask.""We move towards what we persistently ask questions about" (Rossi, 2007).
- *The principle of simultaneity*recognizes that inquiry and change are not separate moments, but occur together. They are interrelated and inseparable. Inquiry is intervention. The seeds of change, the things people think and talk about, the things people discover and learn, the things that provide new knowledge, dialogue and the images of the future that inspire us are within the first questions we ask. The questions we ask set the stage for what we find out and what we discover becomes the stories out of which the future is conceived and constructed.

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- *The poetic principle*states that human organizations, including communities, are an open book. Their past, present and future are an endless source of learning, inspiration and interpretation. Al chooses to focus on the positive aspects of life of communities.
- *The anticipatory principle*states that current behaviour is guided by images of the future. People's expectations are powerful in bringing the future into the present. Communities exist because, the people who govern and maintain them share a vision, dreams and aspirations of what the organization is, how it will function and what it is likely to become.
- *The positive principle*states that momentum for change, requires positive thinking and social bonding-qualities like hope, inspiration and joy in creating one another. If development practitioners use positive questions to guide community development they will achieve more long lasting and effective changes.
- *Every system works to some degree.*It is important therefore to seek out the positive, life-giving forces and appreciate the "best of what is."
- *Knowledge generated by inquiry should be applicable.*It should be relevant and possible to utilize.
- *Systems are capable of becoming more than they are.*Communities can learn how to guide their own evolution by focusing on "what might be."

Advantages of AI

- People Participate in self This raises their self esteem as they recognize their power in understanding self.
- AI reveals common ground of shared values, dreams and aspirations.
- Peoples' hopes and motivation areraised. Al reveals exciting and desirable possibilities.
- People appreciate themselves and their achievements. They identify shared values, culture They take time to celebrate their achievements and successes.
- People's ability to recognize and affirm the positive is strengthened.
- There is likelihood of sustainability of initiatives developed through Al as people are h They use their past successes, victories and achievements as the foundation for their future. Their past successes and achievements make them believe they will also succeed future.
- A culture of continual learning is created.
- Group energy, hope, motivation and commitment are renewed.
- Curiosity and wonder increase creativity and innovations increase.
- There is improvement in interpersonal relationships and ability to resolve conflicts.

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- There is acceptance and inclusion of all levels of society in decision-making.

Challenges

- AI could paint too glorious a picture of the community even when there are glaring problems
- People who are demoralized might not be easily motivated to glorify the past and dream of better future.
- Leadership might not create a friendly environment for AI.
- The community might lack commitment to a long term process to go through the cycle of AI.

Activity

- Define the term appreciative inquiry.
- Describe the steps in AI.
- What are the benefits of using AI in community development?
- What are limitations/challenges of AI?

HOW TO DESIGN A RESERCH STUDY

The design of the research project consists of the stages that a researcher goes through from the time of identifying the project to the writing of the research report.

Major Steps in Research

The following are the major steps in research.

- Selection of the problem
- Development of the proposal
- Review of literature
- Formulation of goal, objectives, research questions or hypotheses
- Design of research project-deciding on research design, research site, sample, instruments to collect data and analyzing data.
- Organization of data
- Analysis of data
- Interpretation of data
- Developing conclusions/recommendations
- Writing the research report
- Publication
- Dissemination of findings

Each of the steps is elaborated in the chart that follows.

A flow chart of major steps in research

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Selection of the problem

Every research starts with the researcher identify a problem whose solution she considers will offer useful information for child development or the education system. A researcher needs to narrow her problem down to address problems related to her discipline and area of interest. For example, if the researcher is in ECD, she should focus on problems related to ERCD. If she is in primary education, she shouldfocus in that area.

Possible Sources of Research Problem / Topics

- Curriculum – types of curriculum and curriculum materials used in schools programmes; process used to develop ECDE curriculum; availability ofprimary school syllabuses and other learning material e.g. books, learning and play materials etc. in ECDE centres; how well the teachers are able I use curriculum materials (syllabuses, text books, learning and play materials).
- Educational materials/learning and play materials- types of learning and play materials available for young children both at home and in school; how these materials are developed; how we they help the children to acquire the concepts, knowledge, skills and attitudes.
- Teaching-learning methods- types of methods used in ECDE centres Primary school; how well they help the children to acquire knowledge, skills and attitudes.
- Assessment tools used to assess children in primary schools.
- Services available in primary schools- types of services; ages of children served etc.
- Children with special needs- types of children with special needs; services available for their both in primary schools and at home; awareness of parents and communities on the needs c children with special needs; types of support given to parents and teachers caring for these children.
- Management, administration and supervision of educational programmes- types of management and administration in schools e.g. types of committees and their roles; other actors etc
- How children learn.
- Methods used to help children learn
- Children's growth and development- milestones of growth and development; abilities of children.
- Child health and nutrition- common childhood diseases; types of foods given to young children, weaning foods, practices and taboos, care of young children, caregivers and their roles etc.

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- Community involvement in the development of care and educational programmes.

Principles in selecting a research problem

The selection of a research problem is a difficult task. It often takes a long time. The following are me principles or considerations that can guide a researcher when selecting a research problem:

- The problem should be related to the researcher's area of study, discipline and interest. This ensures that the researcher has adequate knowledge and skills to carry out the study. She is also able to understand better the literature she reviews, what data to collect, how to analyze and interpret data collected.
- The problem will add new knowledge and learning. The researcher needs to ensure that the problem selected is bringing in new information that will be useful and that will contribute to new learning in the area of study.
- The problem should be her original work. The researcher needs to ensure that the problem she has selected has not been copied from other researches. In case of replicating research, the researcher needs to make this clear. She should however, use different environments and sample.
- This involves the researcher ensuring that:
- Resources are adequate to carry out the study. The resources include money, equipment and people.
- Time available is adequate to carry out the whole study.
- She is able to get adequate sample from the population available within the research site.
- The methodology identified for the study is manageable and easily understood. Her instruments should be simple to understand, easy to administer and analyze. The research site should also be easily accessible throughout the period of data collection.
- Critical mass. The problem should be of sufficient magnitude. It should be a serious problem that affects a large number of people and in a large area. It must have such serious effects on those affected that they feel they must have it addressed urgently. The problem should also have enough variables to studied. There should also be adequate sources of information for literature review.
- Theoretical value. The researcher needs to ensure that the study will contribute new knowledge that will add value to the discipline of study. The research findings should also add more literature for future studies, improve personal and professional growth of the researcher and be useful to other professionals.

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- Practical value. The researcher needs to ensure that the study will provide new knowledge which will help to improve practices of professionals and practitioners and also improve the lives of the people affected by the problem.

How to find a research problem/topic

In order to find a research problem, the researcher needs to be motivated and interested to do research. She should appreciate the value of research in improving her personal and professional life and in contributing new knowledge to the discipline. To be able to get a good research problem the researcher needs to:

- Become a scholar. She must be interested in reading about issues and problems in her area of interest.
- Look for gaps, difficulties and explanations given on issues. She needs to find out what areas have not yet been researched and which areas are lacking data.
- Watch out for inconsistencies and contradictions on issues.
- Observe the problems that emerge during the implementation of projects in her area of interest. What problems have emerged that are of concern to professionals, practitioners, beneficiaries and researchers?
- Read, Listen, discuss and think critically. She must acquire an attitude of reading and questioning whatever she reads; she needs to listen to people and question whatever they say. It is important to start to question the obvious beliefs and practices. Why are they the way they are? Can they be improved? Can they be changed? If, so how? She needs to carry a notebook to write down ideas as they come.
- To criticize and challenge statements made in articles, books, research studies and in what people say.
- To associate with researchers, professionals and intellectually competent people in her field study. This will motivate her to carry out research.
- To follow up ideas that stem from current researches. In some cases it is possible to replicate research done by someone else in order to try and establish whether she will get similar find in other environments (verify findings).

Questions to guide in selection of research problems

The following are the questions that will help the researcher to decide on the research problem

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- Am I equipped to carry out this study? Do I have the required knowledge and skills? Or can equip myself to carry out this study?
- Is this topic a researchable one?
- Can I carry out this study within the time available to me?
- Do I have adequate resources (money, human resources etc.) required to carry out this study.
- Are there enough reference materials for literature review?
- Is there adequate sample from the research site to collect data from?
- What type of data shall I collect? Will I be able to get all the data I require?
- From whom shall I collect data?
- Who will collect data? Will those collecting the data have the necessary knowledge and ski If not, what training do they require? Who will give this training?
- How and when will the data be collected?
- How shall I analyze the data?
- Are the techniques required to analyze my data developed enough to enable me to analyze it to make interpretations, conclusions and recommendations?

SAMPLE AND SAMPLING PROCEDURES.

DECIDING ON THE POPULATION SAMPLE AND ON SAMPLING METHOD

When concluding a research, one must often use a sample of population as opposed to using the entire population. A population can be defined as any set or persons/subjects having a common observable characteristics. It is the total number of subject/person with similar characteristics under the study (research). Sample is a subset or small group of subjects/persons representing the whole population in the research study. The characteristics of a population are called a parameter.

Sampling is a process of selecting a number of individual or subjects from a population such that the selected group contains elements representative of the characteristics found in the entire population.

Reason for Sampling

Why do we sample instead of using the population as the focus of study. There are at least four major reasons to sample.

- First it is usually too costly to do research with an entire population most researchers do not have the amount of money required to study an entire population.
- The second reason to sample is that it may be impossible to research on the entire population. For example, let us say that we want to test the

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mental performance of babies at birth. There are too many mothers who do not make it to clinics and hospitals to give birth let alone testing them.

- The third reason to s ample is that research on the entire population often produces errors. Thus, sampling may be more accurate.
- The other reason to sample is that it may be the only practical way of collecting data, particularly when the population is infinite or extremely large.

Sampling Method

Ideally, in sampling we would like to obtain a sample that is will be representative of the target population. To be representative means to provide a close approximation of certain characteristics of a target group.

The quality of a sample must be judged in terms of the procedure that produces it, that is in terms of its sampling methods/design.

Sampling methods/designs are classified into two (2) broad classes:

1. Probability sampling/Random.
2. Non-probability sampling.

TYPES OF PROBABILITY SAMPLING

1. Simple Random Sampling
2. Interval or Systematic sampling
3. Stratified Sampling
4. Cluster or Multi-stage Sampling

Simple Random Sampling

The usual definition of simple random sample is that it a procedure in which all the individuals in the defined population have an equal and independent chance of being selected as a member of sample. By independent it means that the selection of one individual does not affect any way the selection of any other individual. A more precise definition of simple random sample is that it is a process of selection from a population that provides every sample of a given size an equal probability of being selected.

The main purpose of using random sampling technique is that random sample yield research data that can be generalized to a larger population within margins of error that can be determined statistically. Random sampling is also preferable because it permits the researcher to make certain references about population value e.g. mean, SD, correlation coefficient on the basis of obtained sample value. Random sampling therefore simply means accidental. According simple random is a sampling procedure which provides equal opportunity of selection for each element of the population. There are several techniques of selecting by simple random selection namely; tossing a coin or dice, lottery technique, using random number table.

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Tossing a coin or dice depends on the selection of variables i.e. tail or head for the coin.

Battery technique- This is where a symbol for each unit of the population is written on small pieces of equal sizes of same colour placed in a container, mixed well, and the lucky number (s) drawn (money or box style). The symbol for each unit of the population can be names of participants or symbol yes/no written on identical pieces of papers. One needs to establish the sample size (if only one is needed out of it, we allocate one yes and three (3) Nos).

Random number table. This is more sophisticated methods, particularly useful for large population. Generally, the table consists of a long series of five digit numbers generated randomly by computers. To use the random number table, we randomly select a row or column as a starting point, then select all the numbers that follow in that row or column, depending on the size of the population. Each member of the population is assigned numbers, then as many numbers as possible comprise the sample size is selected from the table. Generally, this table consists of big series of five digit numbers generated randomly by computer. To use the random table, one will need to know the size of the population and sample size and assign a number to each case or population. To start in the table, one needs to randomly select a row or column as a starting point, then select all the numbers in that row or column. If more numbers are needed, proceed to the next row or column until enough numbers have been selected to make up the desired sample size. In effect, you may start at nay random point in the table and select numbers from column, row or diagonally if you wish.

Example one (1).

If there is a population to members and 70 are to be selected at random each of the 70 members is assigned number 1 to 70. The 1st ten members that appear, whenever one begins random, number table, determine the 10 sample member. Since there is only 70 members in the population, two digits random are used. In summary, simple random sampling is a powerful technique for selecting a sample that is representative of a larger population. Nevertheless, it is rarely possible to study a simple random that is perfect. Even if a simple random sample is initially selected, some subject will probably refuse to co-operate and other lost through alteration hence leaving a sample that is not truly random.

Systematic or Interval Sampling

The technique is based on the selection of element at equal interval, starting with a randomly selected element on the population list. The use of systematic sampling is quite common in educational research where large populations are studied and alphabetical or possibly other list of the population member is available.

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Systematic sampling is a procedure by which the selection of the 1st sample member determines the entire sample. The population members that is, their names or type of identification are in some type of order for example by names of pupils in a class may be placed in alphabetical order on the register. The sample size is chosen and the sampling constant 'K' is determined, if sampling constant K = $^{50}/_{10}$ = 5. Hence, every K or the 5th student is selected. K would have a range of (1-) pupil on the last, one would need to randomly select whether to start with the 1st, 2nd, 3rd, 4th, or 5rth pupils. If you randomly select the 2nd pupil, then selection would include (2nd, 7th, 12th, 22nd, 27th, 32nd, 37th, 42nd and 47th pupils). If the starting pupil was number 5, then the selection order would be 5,10,25,20,25,30,35,40,45,50.

The systematic sampling consists of selection of every K. The sampling interval is merely the ration of the number of cases in the pop in the desired sample size. Random startrefer to the process of using a table of random numbers or some other devise to select at random the initial case between 1 and K.

Stratified Random Sampling

In many educational studies, the sample should be selected in such a way that you are assured that certain sub-groups in a pop will be represented in the sample in proportion to their numbers in the pop itself. Such sub samples are usually referred to as stratified sample.

In stratified random sampling, the population is first sub-divided into two or more mutually exclusive segments called strata, based on categories of one or combination or relevant variables. Simple random samples are then drawn from each stratum and then these sub samples are joined to form the complete stratified sample.

Example

Assume that a researcher wanted a sample of 100 pupils from sample of 1000 of the River Bank Academy. It is expected that the student will vary by gender and according to class (year of study).

Allocation of sample size among strata. The decision must be made as to the number that is allocation) that will be selected from each stratum for the sample. There are three (3) methods of allocated:

1. Equal allocation
2. Proportional allocation

- Optimum allocation

The equal allocation are used when all the strata contribute equal number of members or the study.

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A commonly used method is proportional allocation, whereby each stratum contributes to the sample a number that is proportional to its size in the population. The sampling fraction f = $^{sample}/_{fraction}$ which is f = $^{n}/_{N.}$

The third method of allocation is optimum allocation. In optimum allocation, the strata contributions to the sample are proportional to the product of the strata, population sizes and variability of the independent variables within strata. Larger strata with larger variability will have the larger allocation to the sample. This is rarely used.

Summary of the three types of allocation in stratified random sampling

TYPE	**CHARACTERISTICS**
Equal Allocation	All strata contribute the same number to the sample. There are 166 strata, each contribute n/k member to the sample. The sampling fraction varies among strata
Proportion Allocation	Sample allocation is proportional to the strata population size. The sampling fraction is constant for all strata and equal n/N. the larger the stratum, the more member it contributes to the sample.
Optimum Allocation	Sample allocation is proportional to the product of the strata population size and variability. The larger and more variables the stratum, the greater will be the contribution to the sample. The sampling fraction varies the stratums.

In the following example, we shall use the proportional allocation of sample. The River Bankpupils distribution by gender and class in the pre-school.

CLASS	**FEMALE**	**MALE**
KG 1	2000	1500
KG 2	2500	2000
Total	4500	3500

A random sample of 500 pupils using proportional allocation is drawn from 800 pupils taking into account the difference in the class and gender, since the different strata will in the population, the size calculated is proportional to the total population. The sampling structure is the ratio of sample size to population size.

Sampling fraction f= n = size of sample
N = size of population
F = $\underline{n}$= $\underline{50}$ = $\underline{1}$

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N 800 16

For a single random sample, the sampling fractions equal the probability of any member of the population being selected for the sample. Therefore, each category of the population being selected for the sample. Therefore, each category of the population has to be multiplied by this fraction to obtain the corresponding category of the sample. The result is as follow.

CLASS	FEMALE SAMPLE	MALE SAMPLE
KG 1	2000 x 1/16 = 125	1500 x 1/15 = 94
KG2	2500 x1/16 = 156	2000 x 1/16 = 125
	128	219

The total size of the sample becomes 281 + 219 = 500

The next step is to use sample random or systematic sample method to draw the difference from the nominal roll i.e. sample of 125 female from KG 1 and 156 female from KG 2 and 994 males fro KG 1 and 125 from KG 2 making a total of 500.

NOTE: Stratified sampling guard against wild samples and ensure that no sub population will be omitted from the sample. It is also avoids overloading in certain sub populations. Simple random samples have a tendency to distribute themselves according to the population proportion and stratified random sampling with proportion allocation will build this proportionally into sample.

Therefore, proportion allocation in stratified random sampling distributes the sample in such a way that the sampling fraction is the same for all strata.

Multi-Stage Cluster Sampling

In cluster sampling, the unit of sampling is not individual but lather natural occurring group of individual. Cluster sampling is used it is more feasible or convenient to select group of individuals from a defined population. This situation occurs when it is either impartial or impossible to obtain a list of members of the accessible population.

Multi-stage cluster sampling is variant of cluster sampling. The principle underlying multi stage sampling is to start by sampling population which is much more general than the final one. The common multi stage cluster sampling is done when an entire population is being studied involves area sampling as the first step. Area sampling rests on the simple (if not always

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correct) assumption that people live somewhere. If areas are sampled, then anyone has an opportunity to fall into the sample. The researcher 1st draws a sample of designated areas, perhaps city blocks or rural location. Each household in the sampled area is listed and from that list is drawn a final sample.

NON-PROBABILITY SAMPLING

Non- probability sampling refers to the process of case selection other than random. Without random selection, non-probability samples have two (2) weaknesses.

1. The do not control investigators' biasness in the selection of units.
2. Their pattern of variability cannot be predicted from probability sampling theory. This makes it impossible to calculate sampling error or to estimate sample precision. However, while one should be ever mindful of these weaknesses, it would be a mistake to rule out non probability sampling. Many instances, this form of sampling is either more appropriate or practical than probability sampling or the only mean of case selection.

When is Non Probability Sampling Most Appropriate

- In situations where very few cases can be included in the sample, e.g. an investigator doing an intensive study in which unit of analysis is a city, or a particular school, the expense of studying more than one or a few cases, generalizing from sample to population essentially becomes a matter of judgment.

- When studying past events, the archaeologists or historian or educationist often find only a fraction of relevant materials available or accessible. Similarly in contemporary societies, certain individuals or institutions may refuse to co-operate in an inquiry. Under these circumstances, the researcher may either accept a non probability method of case selection.

- In explanatory or the early stages of investigating a problem, when the objective is to become more informed about the problem itself, probability sampling simply may be unnecessary. It will suffice to select a range of cases non randomly without conses for precise statistical generalizations.

- If the population itself contains few cases (pregnancy and dropout) or if an adequate sampling formula cannot be obtained or constructed, then there is no point in considering probability sampling formula. With small population (say less than 20 as in the female primary school teachers in

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the entire Garissa District). Each case should be studied in its own right in comparison with all others. If the population is unknown or readily unidentifiable, as in educational studies of truancy, then sampling generally will consist of studying any and all identifiable and cooperative units.

Types of Non- Probability Sampling
It includes

1. Convenient sampling
2. Purposive sampling
- Quota sampling
1. Dimensional sampling
2. Snow ball sampling

Convenient Sampling

This type of sampling (also called haphazard fortuitous and accidental sampling, the researcher simply selects a required number from cases that are conveniently available. In the case of coast star parents attitude towards cost sharing arrangements towards tuition holiday. The researcher might decide to:

Ask teacher for permission to administer questionnaire to all parents in his school.
Interview however comes in the office or takes his/her child when the teacher (researcher) happens to be there.

Find a convenient spot in the school office or compound from which to administer the questionnaire or interview the parents concerned. TV station and newspaper wanting to tap public opinion in specific issues may interview conveniently available commuters, shoppers, store clerk and others. Such of a case selection are easy, and inexpensive. If the researcher is at an early stage and generalizations is not an issue, then they may be perfectly appropriate. However, convenience sampling is a matter of catch-as-catch can. There is no way of determining to whom, other than the sample itself, the results apply.

Purpose Sampling

In this form, the investigator relies on his or her expertise or expert judgment to select units that are representative or typical of the population. The general strategy is to identify important sources of variations or criteria in the population and then select a sample that reflects this variation (type of school, boarding status and sex). One might select a single unit or sub-population that

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is thought to be typical of the population in important respects or select a few units that correspond to key population differences. In other words, purposive sampling is hand picking the cases to be included in the sample in basis of one judgment of their typically. The subjects area chosen according to a certain specific criteria e.g. Boys and girls school, rural vs. urban residence etc.

Quota Sampling

Quota sampling is a form of purposive sampling that bears a superficial resembling to proportionate stratified sampling. Like the latter, quota sampling begin by dividing the population into relevant strata such as age, geographical region or years. The total sample is allocated among the strata in direct proportion to their estimated or actual size in the population. Finally, to obtain the correct proportions to their estimated or actual size in the population. Finally, to obtain the correct proportions in the sample, interviewers are asked to speak to a fixed quota of respondents in each stratum. So many men and so many women, those of a given age or income and so on are selected. To fill the quota, interviewers are free to choose anyone who meets the quota requirements.

The differences between quota and stratified sampling lies precisely in how cases are selected once quota has been set. In stratified quota sampling, the requisite number of cases within each stratum must be drawn by simple random sampling. But in quota sampling, the quota of cases within the stratum may be filled in whatever vary the investigator chooses.

The potential biases in such procedure are obvious: friends are likely to resemble closely the interviewer themselves; people who visit particular centre may differ sharply from those who do not; and preference for a nicer neighbourhood and homes will almost certainly create a strong socio-economic bias. Thus, representative quota in some characteristic does not ensure representative in others.

Dimensional Sampling

A further refinement of quota sampling where you identify various factors of interacts in a population and obtaining at least one respondent for every combination of these factors.

Snow ball Sampling

After identifying a small number of individuals who have the required characteristics, you use them as informants to identify others who qualify for

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inclusion. The study n drop out or cause of adolescent pregnancies would use this method more successfully.

SELECTING APPROPRIATE RESEARCH METHOD

Here it is a matter of deciding which research designs are appropriated to the study in relation to topic, objectives and data collection method. There are no strict rules for making decisions about what research approach and methods to use. For each study, data collection options and strategies will depend on what kind of information is for and how the information is to be used. Qualitative and quantitative approaches to research are complimentary, and where appropriate should be combined in such a way as to maximize the strength and minimize the limitations.

SELECTING AND DEVELOPING APPROPRIATE RESEARCH TOOL/INSTRUMENT;

In education, as in other behavioural sciences, researchers rely mostly on reactive research method as opposed to unconstructive methods. In other words, in much educational research, the researched person (respondent) is aware of being studied and reacts to stimuli, or the instrument of studying them in the form of the questions presented by the researcher. The most frequently used tools of gathering information is by directly asking respondents to express their views or opinion.

Data Collection Tools/Instruments

They include

1. Questionnaires
2. Interviews
3. Observational guidelines
4. Rating scales
5. Ranking order

Critical Considerations when Constructing Research Instrument/Tools

The critical consideration while construction instruments are:

Objective of the study

A researcher must have a clear understanding of what to obtain from the results of the study. This means that the researcher must be able to anticipate and spell out the type of information needed. First, the researcher must have a clear vision of the time frame for the research specification whether or not to focus current educations practices or trend over time (longitudinal studies).

Type of population/sample:

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The research should be aware that some type of instrument are unsuitable to some groups of people / respondents depending on their literacy level, and type of work or professional culture and level of socialization . For illiterate population questionnaire will be unsuitable, hence of interview or observation method will be necessary.

Geographical Distribution of subjects

A researcher needs to be conversant with geographical location and spread of the sources of information before making decision on the type of instrument to be constructed and administered. One need whether the study will survey the whole country (census).Selected provinces, district, division or location. The span of the will dictate the type of instrument to be used.

Nature of Research Item.

In the special cases, each item must be developed to measure a specific aspect of the stated objectives. The item (whether in multiple or open ended format) should reflected why the question is being asked and how the item will be analyzed. The type of item should be able to be understood and interpreted by the respondents. It is usually desirable for the researcher to deve3lop during table to assist easier analysis later.

Questionnaires

A questionnaire is an instrument used to gather data, which allows a measurement for or against particular view point. A questionnaire has the ability to collect a large amount of information in a reasonably quick space of time.

This is a big advantage but do not let this deceive you into thinking that questionnaire design is a straight forward operation. Asking correct question about a particular topic need careful planning.

Planning checklist.

In planning a questionnaire needs careful consideration. The checklist below should be constructed before moving on actual design. You should be able to answer all these questions in a positive way that should be able to state the specific goals of your study.

When you look at the goals of the study do they convey a picture of something that is worth doing?

Be sure in your mind that the worth is not of trivial nature.

Do you know enough of subject area to create a questionnaire that has inclusive items?

How much do you know about the characteristics of your target population?

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What is the approximate cost of your study and how long will it take to complete.
What sort of sample are you going to use.

NOTE: A common problem with many questionnaires is the number of irrelevant items or questions that are included.

Take a look into a few questionnaires. What are the most common items you see at the beginning of these questionnaires? Age gender, marital status, religion, occupation are families items contained in question. Justification for inclusive is all important and the contents must reflect the aim and objective of your study. If marital status is included in the study of pre primary education performance, why should you include it. Or if there is no sex difference in performance of certain cognitive aspect in pre primary pupils, it is not necessary to include it in the questionnaire.

Question Types

Question con come in a variety of shapes but in general there are 4 types:-
Background questions
Closed-end questions
Partially closed end questions
Open end question

Background Questions

This type of questions is used to gather demographic details from the group being studied. It is quite common to find background question quite early in a questionnaire because usually they are fairly easy to answer.

Please provide the following
Gender……
Teaching Experience :……
Marital status :……
For example:

NOTE: Only seek background information that is relevant to your study and its objective.

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Close-End Question

This type of question is used to collect information or determine opinion on issues by giving the respondent a set of choice or response options.

For examples: – 1

Example 2:

The teaching example has unordered responses. Choice are described but there is no single dimension related to the options. Each of the 6 categories has to be viewed as independent of the others.

Partially closed

Partially closed End questions with this type of questions the respondent also has option but is not forced to select of them. There is provision for other reasons to be listed, once which we include in the fixed choices.

For example

As an teacher one of the following areas would you specialized training in (tick one box only.)

1. School administration and management degree
2. food and nutrition degree.
3. Degree in curriculum design in

Another area, please describe ……..

Open – end question

These are sometime called free response types because they allow the respondents to answer in their own words. Open questions are often used to explore issues.

Question construction

Constructing good questions may take some time and what first may look as ideal can, after a pilot test, be shown to be unreliable or infective. Below are some useful hints on questionnaire construction.

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1. Specific questions will produce specific answers: try to avoid vague sounding questions. They are usually difficult to answer. For example: have you read the book recently? Different people will interpret **recently** in various ways. Substitute recently with, **within the last 7 days or month.**
2. Over precise questions can cause problems: if you make the question too difficult to answer. Example: how many of your classmates did you talk to yesterday?
3. Keep the language simple: look at who will be answering your questions and write at their level. Avoid jargon or technical terms.eg Donor conditionality,jittygritty,without much ado.
4. Avoid the use of double negative: two negative in a question can be confusing and the purpose is to facilitate a clear response. Example: lack of midterm exams is not something to worry about.
5. Limit each question to a single idea. Example : are you favoring of reducing the number of teaching hours and increasing pupil intake on the KG level.What will be the answer to the above question tell you? (Yes or no).
6. Avoid emotional charged words: such words can create bias and introduce a highly personal tone which many respondents may feel uneasy with. Do you usually abscord Friday afternoon lessons? .How is your relationship with your headteacher?
7. Avoid leading questions there are questions which by their wording suggests the answer. Example: do you think we should increase pre-primary academic staff pay since they work such long hours and provide available services?, Do not think learners who fail in one class should be told to repeat to avoid failure in their K.C.P.E?, Don't you think the current wrangles over Mau forest will split the ODM?
8. Be careful of personalization: personalization can cause problems if you are seeking opinion of a general nature. Example: what do you think of day care services? Don't you think Kamba people practice witchcraft?
9. Try to avoid subjective terms: such as "usually", "often" and "many", as they can be interpreted in various ways. For example "many people" to a city dweller will probably mean something different to someone in living in rural isolation as far as population of their setting is concerned.
10. Keep the item as short as possible: there is nothing more irritating than a lengthy sentence that needs someone to re-read again and again before he gets the message.

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11. Ensure that all response options are included: it is surprising how many questionnaires do not include categories such as "don't know" or "not applicable". Knowing that respondents don't know can be variable piece of information.
12. Avoid acronyms and abbreviation e.g STD may stand for standard Trunk Dialing or Sexually Transmitted Disease. So use full letters in questionnaire.
13. Pilot the questionnaire: once you have constructed a questionnaire, try it out on a small sample to check if it is operational.
14. Intensity questions: it is quite likely that you may need to measure strength opinion. The most common format is the LIKERT This permits the respondent to select from a number of degree of intensity (usually five). For example.

This type of item is usually associated with the measurement of attitude.

15. Constructing relevant questions: Depending on the subject matter of the questionnaires there may be times when you will have difficulties arriving at appropriate questions i.e. those that will give you the information you are seeking. It is good to think deep and focus in questions that can give relevant information.

<u>The Questionnaire Layout</u>

As with most products the usual impact can be a big plus. Check your question or the following:

Physical layout – It may be restarting the obvious but, make sure your questionnaire is neat, easy to use, easy to code and easy to store.

Production cost – The information you get should be worth the cost of production.

- Identification – put an individual identification number on each questionnaire you distribute.
- Sender's name and address – always put your name and address on the questionnaire just incase the original questionnaire is misplaced.
- Use fairly good quality paper and if possible use only one side.
- Space your questions. Nothing looks worse than a large group of questions compressed into a single page.
- Clear instruction – it is important to have clear instruction on how to answer each question.

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- Clear explanation – if your questionnaire has different section then begin each one with a few words of explanation.

Question

What are the advantages and disadvantages of a questionnaire as a data collection tool?(10mks).

Interviews

Interviews involve a researcher meet a respondent physically or through phone and series of questions orally asked as part of research project. There are 3 types of interviews in an education set up, especially in primary program:

1. Unstructured interview
2. Semi structured interview
- Structured/systematic interview.

Un-Structured Interview Technique

It is one of the most widely used techniques of data collection in education. In this approach of interviewing, the researcher has some idea in mind of the topic covered and may use some sort of topic list as a reminder, but there is minimal control over the order in which topics are covered and over the respondent answers. In unstructured interviewing, neither the specific questions to be asked nor the range of type of possible answer are pre-defined.

Unstructured interviewing is formal and conversational, and the aim is to get inform unto open up and to express themselves in their own terms.

Strength unstructured interviews:

- The main strength of an informal conversational approach to interviewing is that it allows the interviewer to be responsive to individual differences and situational characteristics.
- This style of interviewing builds well upon observation and can help discover relevant questions and their appropriate wording ensure communication with the respondent.
- Its also useful for building initial rapport with informant, before conducting more formal interviews and for talking with the people or in situations where more formal interview may not be tolerate e.g. in studies of street girls drug addiction in relationship to effects on their pregnancies and their infants
- Unstructured interviews are also useful for studying sensitive topics such as sexuality or certain tetratogen topics experienced by mothers

Weakness of unstructured interview:

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- Because there is no set format for conducting conservational interviews each interview tends to be unique. This makes it difficult to systemize and analyses data
- It may take several conservation before obtaining a similar set of information from each informant thus time wasting

Semi-structured interview

Semi-structured interview area based on the use of interview guide. This is a writer list of questions or topics that need to be covered during the interview. However, the exact order and wording of the questions may vary from respondent to respondent. The interviewer may still follow leads and new topics that arise in the course of the interview, but the interview guide is a set of clear instructions concerning the main question to be asked or topic to be probe. Semi structured interview include: Depth / Focused interviews that intensively investigate a particular topic and case studies that collect comprehensive, systematic and in depth information about particular cases of interest.

Structured systematic interviews

Structured interviewing involves subjecting every informant in a simple to the same stimuli i.e. asking each informant the exact same question. Survey such as are most familiar with the structured questionnaire.

The data collection technique under discussion produces numerical quantifiable data.

Nonetheless, these methods are included in the discussions of qualitative research because their purpose is to help describe and analyses the culture and behavior of respondent from the respondent point of view. Furthermore, their effective user depend on prior understanding of the study of population's view and perspective

OBSERVATION TECHNIQUES AND GUIDELINES

Observations provide information about actual behavior. Direct observation of behavior is useful because some behavior involves habitual routine which people are hardly aware of. In such cases, observation can reveal more information than interviews or questionnaires. Direct observation also allows the research to put behavior in context and thereby understanding it better. Observation can either be unobtrusive and non-reactive or obvious and reactive.

Unobtrusive observation is whereby people are observed without them knowing, but involves serious ethical dilemmas. Here we will only discuss obvious and reactive techniques. The approaches are three, namely:

- Participant observation

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- Unstructured observation
- Structured observation

Participant Observation

Participant observation refers to an approach whereby the investigator becomes to a greater or lesser extent, an active functioning member of the culture under study for example a family health organization, teaching staff or community. The investigator participated in any activities appropriate for a person of the status in which is assumed. Observe what others do and in general attempt to see through the eyes of a member of a culture rather than through those of an outsider. Participant observation in itself does not constitute a specific data, unstructured interview and unstructured observation; rather it simply means living in or hanging around the community or culture under study. In participant observation the people being observed are aware of the researcher's presence in the community and usually know the purpose of the researcher's presence is to learn about their way of life. Participant observation produces textual descriptive account.

Strength of Participant Observation

- Participant observation facilitates all other data collection in the field by helping to reduce reactivity (the degree to which people change their behavior when they know they are being studied). People become more and more comfortable with researcher's presence.

- It helps the researcher to learn how to formulate appropriate and relevant question in the native language.

Structured observation

In structured observation, the observer is an on-looker, he or she observes as an outside. Those observed know they are being observed but usually do not know the exact behaviors that are being observed and recorded. The focus tends to be on small number of very specified behavior. Only those behaviors appearing on a pre defined observation list are recorded. Structure observations produce quantitative data with little contextual description. In order to conduct structured observation, behaviors and actors of interested must be clearly specified.

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The importance of the timing of observations must be determined (time of day, season e.t.c.) and focal behaviors must be defined, broken down so that recording is reliable.
This approach ensures that the observation will be optimally reproducible, comparable and quantifiable.

Unless the behaviors of interest are well understood, it is not possible to develop a good observation recording sheet. Therefore structured observations should never be used before first, rather conducting unstructured observations and interviews to explore what really needs to be known, and in what detail.

The approach may include continuous monitoring, spot check and rating checks.

- Spot checks are observations whereby the observer records the presence or absence of behavior or physical characteristics of interests at the first moment of observation. For example, to make a round after lunch to see how many pupils have left dishes you washed after meals (as an indicator of dishwashing behavior).
- Rating checks require the observer to make judgment on individual and environments e.g. "woman washes hand" is a pure observation of behavior while "women's hands are clean" involves judgment by the observer that may not be true. Rating should be made as explicit as possible.
-

Strength of Structured Observation

- They provide precise, numerical result which are applicable to statistical analysis and can be repeated to monitor behavior change over item.

Weakness of Structured Observation

- The problem to be studied need to be clearly defined.
- Training of observer is intensive and time consuming.
- Pre-determined structure of the observation limits discovery of other potentially relevant behaviours.
- It gives the researcher an intuitive understanding of what is happening in a culture; helps the researcher understand the meaning of the data being collected; maximizes the researcher's ability to make valid statement about the culture being studied.

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- It is useful when if the context is new or not willknown and when the topic of interest is especially complex. It is particularly useful at the beginning of research when a problem is not well understood.
- It is useful when the situation of interest is hidden from the public or when people appear to have significantly different than to outsiders.
- It is especially appropriate for understanding processes, events, norms and values and the context of social situation.

Weakness of Participant Observation

- Depending on the population studied, participant observation can be quite time consuming, taking anywhere from a few weeks to many months.

- To conduct participant observation, a researcher must speak the local language well, be skilled at observing the little details of life, have good memory (in order to write down later what was observed) and be skilled at writing what was detailed and copies about what was observed or discussed.

- Sometimes researchers become so familiar with the future that it becomes more and more difficult to notice things that should be considered different or important.

Unstructured Observation

Also referred to as open observation. The observer takes the position of an onlooker rather than a participant observer. Those being observed know that they are being observed, but the "what" to be observed is only very broadly defined for example, infant feeding.
Unstructured observations are broadly focused and their aim is to observe behaviours in its context .Data are collected in form of detailed descriptive accounts. Unstructured observations are often used or conducted when the researcher is trying to discover unknown aspects of a problem. They allow the discovery of "surprises" which can then be followed up in later stages of the research. Unstructured observations are particularly helpful for understanding behaviors in their physical and social context.
Data coding, entry and analysis can be very time consuming.

RATING SCALE

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Scale can be created from any number of concepts or attributes, and items can be rated on a single conceptual scale or each may be rated on a series of scales presenting a variety of concepts or attributes. Scales can be presented numerically. For example:

1. Circle the number that corresponds to the level of severity you will associate with pupils dropout rate in each stage in your school.

Pre primary stage 0 2 3 4 5 6 7 (from least to severe)

Lower primary stage 0 2 3 4 5 6 7 (From L to S)

Upper primary stage 0 2 3 4 5 6 7 (From L to S)

2. Mark an X on the line below indicating where you will rate the drop out rate of pupils by gender.

Boy

Girls

Strength of rating scales

Rating scale are easier to administer.

Weakness of Rating scales

Rating scale can be extremely sensitive to responses bias (the propensity of an individual to always use one end of the scale or a narrow range in the middle of the scale).

RANK ORDER METHODS

Rank order method require information to rank items i.e from most to least in term of a specific characteristic, for example dropout rate by class and gender in terms of severity.

Complete rank ordering method usually require that informant be literate and involves presenting respondent with a list of items which they are asked to order from most to least on some scales or attribute by putting number next to each item. Another method is to present each information with visual stimuli or stack of cords on which items names are written and ask them to order the item

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or cards from most to least (best to worst or whatever the attribute of interests is).

Partial rank ordering: Pair each item with each of the other item. The pair of items are presented to respondents who are asked to indicate which is more or less, a best of worst, most ordering is obtained by summing the number of items each item was chosen.

Strength of Rank Order Method

The complete rank ordering technique produces a great deal of information and is productive for the time spent by the informant. The method is ideal for study individual differences.

Paired comparisons are probably the easiest and most reliable method to use with illiterate when there is small number of items to be ordered.

Weakness of rank order method

The complete rank ordering technique can be tedious for non-literate respondent. For paired comparison, pre testing is crucial for identifying the maximum number of pairs that informant will tolerate. Some researcher has found that even as few as 15 pairs can be tedious.

DATA COLLECTION PROCEDURE PROGRAMME AND ORGANIZATION

The process of data collection is much decided by the research and amount of resources available (time and money).

After choosing appropriate research tools which can be a combination of 2 or more research tool, if not one, a researcher may decide to do it in personal or hire other skilled people in the collection of data.

For interview he may decide to interview informant through the phone or meet them personally for an interview.

As for questionnaire he may present them personally to informants, then collect in personally after the informants have filled the questionnaire or he may send them through post office and wait for informant to fill the questionnaire and send them back to him.

ETHICAL CONSIDERATION IN RESEARCH

Through the process of data collection, the problems of persuading participants to co-operate with the researcher and participate in the research is ever present. Lack of co-operation can be disastrous in a research task. Participants

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have the right to refuse openly to take part. Generally, acceptable ethical right must be followed. This is particularly the case when dealing with public schools and other educational institutions. When you are working with school children and teachers as subject, its necessary to have understanding and co-operation of school administration, (the education officer, head teacher, teacher). Interested community groups and other key stakeholders.

Ethical Principles

Researcher needs to be conversant with ethical principles to ensure that human rights and public relations are strictly adhered to. Apart from one overall observation of professional outlook, mannerism, mien and decorum, one needs to ensure that the following are observed.

1. **Informed Consent**

Educational or social science research invades person's privacy. An investigator might want information of a private nature or observed people in situations harmful to or at least uncomfortable to participants. People should not be subjected to research of such nature unless they agree to participate in it. Participation in research must be voluntary and people have the right to refuse to divulge certain information about themselves. This right to privacy demands that direct consent for participation must be obtained from adults and incase of children from their parents or teachers.

2. **Ensure confidentiality**

The participant must be assured that the information will be kept confidential and only used for used for the purpose of the research alone. The participants must be told who will access to the data. In this case, the participant must be assured that the data will only be used for the stated purpose of the research and that no undesirable person will have access to the data. Assured of these conditions, a respondent will feel free to give honest and complete information. Therefore to ensure confidentiality, researcher should:

- Not include or write names of respondent to the data except a code
- Ensure nobody can link the data to specific subject.
- \substitute names with number so that only a person who has access to the codes can identify the subject.

3. **Plagiarism**

This is the use of somebody else ideas or research purporting it to be yours, without acknowledging the source. It is an academic crime and it is punishable. It is protected by writers' copyright (patent rights to writers' creation). When using somebody else ideas, you should quote the worker's work using quotations.

4. **Risk to Participants**

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The subject/participant should not be harmed in anyway – mentally, psychologically, physically, emotionally or morally for the sake of obtaining information in the name of scientifically research. If the research involves any risk, the subject should be informed. If school children are involved in the study for example, the parent or guardians should be informed and written permission secured from them.

5. **Permission to interview children.**

In addition to parental consent, researcher should seek the informed consent of children age seven (7) and above for research participation. The use of deception in research with children is especially risky since debriefing can undermine their basic faith in the trustworthiness of adults.

6. **Using of human beings as specimen object.**

At no one time in a research should human beings be subjected as specimen in the study i.e. being used without human integrity or reduced to animal level.

7.

Since studies can only take effect when some vital information is withheld or subject not told, the absolute truth is necessary. Although needed data can be collected without deception, practical considerations such as limited fund and time lead some researchers to use deception. This is used in extreme cases where serious issues such as crime and potential respondents may be violent if they discover that one is collecting data on them (e.g. criminals).

8. Should ensure good human relation for example, it should not bring conflict between Head teacher and his teacher or among pupils or in families and societies.

MONITORING AND EVALUATION

MONITORING.

Monitoring is the systematic and continuous assessment of the progress of a piece of program/work over time. It is a basic and a universal management tool for identifying strength and weaknesses in a programme.

EVALUATION.

An evaluation is the assessment at one point in a time of the impact of a piece of work/programme and the extent to which stated objectives have been achieved.

Importance of monitoring and evaluation

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1. They help to show what impact the work/programme is having and the progress towards meeting objectives so that activity can be adjusted if necessary.
2. Monitoring and evaluation assist in maintaining high standard of a programme i.e. assess the quality, effectiveness and efficiency of activities, as well as the volume of work.

- They make sure resources are used effectively i.e. to find out what resources are required to produce certain effects, or how resources can be distributed differently to be more effective.

1. They are used to plan work i.e. to show what and who will be available when, and how work could be affected by seasonal trends.
2. Evaluation and monitoring identify problems and find solutions at nearly stage of the programme to make sure problems can be discussed and tackled before they become too serious.
3. They help staff feel their work has a definite purpose. Knowing more about objectives, progress, impact and quality of work in a programme will help staff feel motivated and involved in the programme.

- They help analyses the style of work, whether this is the best way of working to achieve more self – motivation, capacity building awareness of gender in equality etc.

Monitoring and Evaluation tools/Method

There are many different methods of data collection. The most appropriate method can be selected according to the kind of information that is needed and when and how it will be used. Combination of objects should always be used so that data can be cross checked.

Possible method for collecting Data at specific Point during monitoring

1. **Surveying with a set of questionnaire**

Survey samples can be selected to compare population affected by the programme or to compare current data with the results of a baseline carried out before the programme started. Regular surveys can be used in a monitoring system to collect information about key indicators to see how the target population is affected by a piece of work over time.

2. **Participatory method**

Participatory method including those discussed in PLA can be used for finding out how the different people involved in a programme, including people affected by the programme, views its progress.

3. **Measuring skills and knowledge**

The skills and knowledge of trainees in ECDE programmes can be assessed to measure effectiveness of activities designed to train and teach people or to

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influence their attitude or behaviour method. Measuring can be done through the following methods-:

- Written, oral or practical test
- Role play and games
- Demonstration of a particular task
- Observation of normal practices using checklist

Possible methods for collecting data regularly for monitoring system include-:

1. **Regular record-keeping: Forms and diaries**

Some information about activities is recorded on regular basis. Forms and set formats are often used for recording quantitative and qualitative information.

The following factors are important

- Good form design will facilitate the accurate recording of information
- Clear instructions on how to use form should always be available.
- The form should contain enough information to be useful to the people collecting the data.
- The people collecting the data need to understand how the information will be used.
- The information needed to complete the forms should be available without too much extra effort.
- The format should be the same in the different forms and registers used.

1. **Supervision checklist and Reports**

The collection of data during supervision meetings provides an opportunity to discuss the information as it is being collected. This can be useful for both supervisors and those being supervised. Checklist and set formats can be used to ensure that information about indicators is collected.

- **Case studies**

Case study can be used to examine the impact of a programme on a particular set up (community). A checklist can be used to ensure certain questions are addressed, without being restricted only to a pre-determined indicator. It can be useful way to looking at unexpected outcome and indirect effects of a programme and to see how other factors have contributed to any change in people's lives.

1. **Spot checks**

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Periodic studies into a particular aspect of the programme. Selected activities may be monitored in detail over a specific period. For example, all staff could fill in activity sheet every day for several weeks to assess the effectiveness of their use of time. This can be useful exercise leading up to a more formal evaluation, since it gives staff and participants a clearer idea of what they are doing, especially if they do not usually keep regular record of activities.

Uses of monitoring and evaluation data

The data collected is used:

1. To assess the progress and impact of programme toward achieving set objectives.
2. To clarify the objectives of the programme that has been running for some time.
3. To identify the issues and problems faced by the programmes, so that they can be

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